

Exact mixed relations in the original arithmetic observation kernel

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The Split-Zero programme represents finite zero-divisor classes of the completed zeta function by polynomials in an arithmetic theta-source Hilbert space. The original observation map leaves a kernel. Its metric is obtained by minimizing over all source representatives before restricting to that kernel.

This edition gives an exact finite determinant formula for that metric. It retains every mixed relation, the source mass, and the original coefficient frame. The formula identifies the remaining large-degree kq calculation; it does not evaluate that coefficient or prove an RH conclusion.

Reading guide. MR1–23 gives the new calculation. The following sections retain the complete mixed/confluent proof and independent verification, the original conductor frame, and the earlier Gamma, arithmetic-coercivity and joint-minimum proofs on which it depends. The final diagram shows the exact maps and matrix sizes. Human sources are cited where used; programme predecessors remain credited as predecessor results.

The original observation kernel and all mixed relation terms

The finite determinant theorem of Akemann and Vernizzi [1, Th, Ddef, deg] applies to the programme's original polynomial measure. The complete finite proof, including repeated roots and both cross-Gram conventions, is retained in the accompanying derivation [2]. This note calculates its receiver in the actual conductor-defined observation kernel. It does not replace that kernel by the conductor ideal. Every coefficient of the original period and amplitude matrix is retained.

The earlier source–relation volume identity [3, (11)–(16)] and the earlier kernel/observation determinant factorization [5, FW37–40] remain their earlier results. The new calculation expresses the actual restricted determinant as a complete sum of the literature's mixed relation determinants, with the original invariant coefficients and all off-diagonal terms. It also gives an inverse-free finite matrix expression for the Gamma determinant in [6, JM21–28]. No large-degree coefficient is asserted here.

1 The actual source, roots and coefficient frame

Retain the original simple off-line quartet parameterization

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \equiv 1 \pmod{4}, \quad k \geq 9, \quad c = k/2, \quad q = (k+1)^2. \quad (\text{MR1})$$

The period is on the original five-orbit locus of [4]. This is the domain on which that source proves its rank and frame formulas; the period and original units are not assigned new values. Order pairs lexicographically, and keep

$$v_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - v_{ab}). \quad (\text{MR2})$$

These roots are distinct: equality forces separately equality of both indices because $\delta, \gamma > 0$. Let $E = \mathbb{C}[S]/(\chi)$. The original source is

$$d\mu(u) = w_h^{*k}(u) du, \quad w_h(t) = \frac{|(2\xi/h)(1/2 + it)|^2}{2\pi}, \quad S = c + iu. \quad (\text{MR3})$$

All polynomial moments are finite and its polynomial Gram is positive definite, as proved from the theta source and Gamma decay in [3, 2]. Its mass is μ_h^k , not one. The source map is the original $P \mapsto P(D_1 + \cdots + D_k)F_h^{\otimes k}$, where $\mathcal{M}F_h = 2\xi/h$; its full arithmetic value is $\eta[P] = v_h^{\otimes k} P(\sum_i M_{s_i})1$. Thus the original Taylor unit and the full quotient map are unchanged.

Here is the exact coefficient frame from OCF1–24 [4]. Keep $\mathbf{H} = F^{-1}\Pi U\mathcal{E}_{\text{prim}}$, $Q(x) = Q_0(\mathbf{H}^{-1}x)$ and $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. Write $gf(x) = f(D^{-1}x)$, $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and $\mathcal{A} = \prod_{a=1}^4 g^a Q$. Under the original substitution $x = \mathbf{H}(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T$, its restriction is the full biform

$$A = \sum_{a,b=0}^8 a_{ab} z_0^a z_1^{8-a} w_0^b w_1^{8-b} \neq 0. \quad (\text{MR4})$$

Let M_A be multiplication by this actual biform from bidegree $(k-8, k-8)$ to (k, k) , in those same lexicographic coefficient bases. There are $q' = (k-7)^2$ columns. The OCF elimination at the largest nonzero coefficient of A produces a strip insertion S_k , reduction N_k , and independent invariant-image columns Z_k , with

$$N_k S_k = I, \quad \ker N_k = \text{im } M_A, \quad \text{rank } Z_k = 8k - 32. \quad (\text{MR5})$$

Its selected row set I has invertible block $(Z_k)_I$, and T is the complementary strip row set. The original coefficient quotient is

$$\pi = (\text{row}_T - (Z_k)_T (Z_k)_I^{-1} \text{row}_I) N_k, \quad Z = [M_A \ S_k Z_k]. \quad (\text{MR6})$$

The full finite construction and proof of MR5, including all nonzero leading-coefficient and pivot charts, is retained in [4]. In particular these are actual period coefficients, not generic ones.

For precision, MR5 immediately proves that Z has independent columns. If $M_A x + S_k Z_k y = 0$, applying N_k gives $Z_k y = 0$, hence $y = 0$; injection of multiplication by the nonzero biform gives $x = 0$. The OCF invariant image consists exactly of these columns: its reduction is $\text{im } Z_k$, and the conductor is its kernel under N_k . Subtracting $S_k Z_k y$ from a vector with the same reduced image leaves an element of $\text{im } M_A$. Therefore

$$\ker \pi = \text{im } Z, \quad r := \text{rank } Z = q' + 8k - 32 = q - (8k - 16), \quad m := \text{rank } \pi = 8k - 16. \quad (\text{MR7})$$

Surjectivity of π also follows by inserting the T rows in the strip. This proves the ranks and the whole image used below. Keeping only M_A would give a different, larger annihilator of dimension $16k - 48$, not m .

Let $\mathbf{V} : E \rightarrow \mathbb{C}^q$ be the literal root-value map $[P] \mapsto (P(v_{ab}))_{ab}$. It is invertible by monic division and the root Vandermonde determinant. OCF24 identifies the coefficient pairing with the original dual observation pairing: a biform coefficient column z acts by $z^T \mathbf{V}[P]$. Consequently the original kernel $K = \ker \Lambda$ is given exactly by

$$\mathbf{V}K = \ker Z^T = \text{im } \pi^T, \quad I_K = \mathbf{V}^{-1} \pi^T. \quad (\text{MR8})$$

The equality follows from $Z^T \pi^T = 0$ and the proved dimensions. This is a complex-linear dualization, so it uses a transpose, not an adjoint. The adjoints enter only in the physical Gram calculations.

2 Full minimization before restricting the kernel

For $N \geq q - 1$ put $d = N + 1$ and $n = d - q$. Let p_j be the monic orthogonal polynomial in the original S coordinate, with actual norm ω_j . Define

$$K_d(v, \bar{w}) = \sum_{j=0}^{d-1} \frac{p_j(v) \overline{p_j(w)}}{\omega_j}, \quad C_d = [K_d(v_\alpha, \bar{v}_\beta)]_{\alpha, \beta=1}^q, \quad D_d = \prod_{j < d} \omega_j. \quad (\text{MR9})$$

The root order is the order in MR2. Mapping the complete orthogonal source basis to its value columns gives covariance $C_d > 0$. If H_d is the monomial source Gram and E_d its evaluation matrix, then $C_d = E_d H_d^{-1} E_d^*$. For prescribed values y , the polynomial coefficient vector $H_d^{-1} E_d^* C_d^{-1} y$ has value y and is orthogonal to every vector in $\ker E_d$. The squared norm of another lift is its norm plus the squared norm of their difference. Thus the unique minimum has norm $y^* C_d^{-1} y$, including every admitted relation χP , $\deg P < n$. It follows that

$$G_N = \mathbf{V}^* C_d^{-1} \mathbf{V}, \quad H_{K,N} = I_K^* G_N I_K = \bar{\pi} C_d^{-1} \pi^T. \quad (\text{MR10})$$

This proves the receiver in the original metric; no restricted inverse has been substituted for the inverse of the whole covariance.

There is an exact complementary determinant with no inverse in its variable matrix entries. Set

$$g_{\pi, Z} = \frac{\det(\bar{\pi} \pi^T)}{\det(Z^T \bar{Z})} > 0.$$

Then

$$\boxed{\det H_{K,N} = g_{\pi,Z} \frac{\det(Z^T C_d \bar{Z})}{\det C_d}}. \quad (\text{MR11})$$

Here is a complete proof retaining both frame factors. Put $U = \pi^T$ and $V = \bar{Z}$. Their Euclidean cross Gram vanishes because $U^*V = \overline{\pi Z} = 0$. Both have full column rank and their dimensions sum to q . Introduce, only to prove the finite matrix identity, $U_0 = U(U^*U)^{-1/2}$ and $V_0 = V(V^*V)^{-1/2}$. The matrix $[U_0, V_0]$ is unitary. In this matrix's coordinates write C_d as $\begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$. Its upper inverse block is $(A - BD^{-1}B^*)^{-1}$, by block multiplication, and its determinant is $\det D \det(A - BD^{-1}B^*)$, by determinant-one block elimination. Therefore $\det(U_0^* C_d^{-1} U_0) = \det(V_0^* C_d V_0) / \det C_d$. Undoing both changes of basis gives MR11 exactly. This auxiliary unitary matrix is not a new source metric or a changed programme frame.

The four cutoffs use the identical π and Z . Hence, with $\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}$,

$$\mathcal{R} \log \det H_{K,N} = \mathcal{R} \log \det(Z^T C_{N+1} \bar{Z}) - \mathcal{R} \log \det C_{N+1}. \quad (\text{MR12})$$

The constant $g_{\pi,Z}$ cancels because its coefficient is $1 + 1 - 1 - 1$. It was retained in each finite metric first.

3 The complete mixed relation sum

For each increasing r -element subset I of the original q roots, define

$$z_I = \det Z_I, \quad \chi_I(S) = \prod_{\alpha \in I} (S - v_\alpha), \quad V_I = \prod_{\alpha < \beta \in I} (v_\beta - v_\alpha), \quad V_\chi = \prod_{\alpha < \beta} (v_\beta - v_\alpha). \quad (\text{MR13})$$

Subscripts on Z denote row selection in that exact order. For monic relations f, g define the original mixed determinant

$$B_t(f, g) = \det \left[\int S^i f(S) \overline{S^j g(S)} d\mu \right]_{i,j=0}^{t-1}, \quad B_0 = 1. \quad (\text{MR14})$$

It has orientation $B_t(g, f) = \overline{B_t(f, g)}$. The mixed theorem of [1], with its full proof in [2], applied to these r -root subsets gives

$$\det(C_d)_{I,J} = \frac{V_I \bar{V}_J}{D_d} B_{d-r}(\chi_I, \chi_J). \quad (\text{MR15})$$

Indeed its polynomial kernel has exactly d terms, so the integrated relation degree is $d - r$, not $d - q$. Expanding the two determinants of the matrix product in MR11 gives Cauchy–Binet twice:

$$\det(Z^T C_d \bar{Z}) = \sum_{|I|=|J|=r} z_I \bar{z}_J \det(C_d)_{I,J} = \frac{\mathcal{A}_d}{D_d}, \quad (\text{MR16})$$

where every term, including $I \neq J$, is kept in

$$\mathcal{A}_d = \sum_{|I|=|J|=r} z_I \bar{z}_J V_I \bar{V}_J B_{d-r}(\chi_I, \chi_J) > 0. \quad (\text{MR17})$$

The product expansion follows by expanding each determinant into permutations: selections with repeated indices cancel in pairs, and the remaining increasing selections give the displayed signed minors. There are no additional signs because Z_I and both kernel row and column sets use the same

increasing order. Positivity of the whole sum follows from $\mathcal{A}_d = D_d \det(Z^T C_d \overline{Z})$, since $D_d > 0$ and $\overline{Z}$ has independent columns. A single mixed summand is not asserted positive or real.

The full-root case gives $\det C_d = |V_\chi|^2 B_{d-q}(\chi, \chi)/D_d$. Substitution proves

$$\det H_{K,N} = \frac{g_{\pi,Z}}{|V_\chi|^2} \frac{\mathcal{A}_{N+1}}{B_{N+1-q}(\chi, \chi)}. \quad (\text{MR18})$$

This is a ratio for the original kernel, not the determinant of its containing conductor ideal. The numerator's relation determinants have size $N + 1 - r = n + m$; the denominator has size n . At $N = q - 1$ the denominator is exactly $B_0 = 1$ and the numerator sizes are exactly m . No negative degree occurs. The associated four-return is

$$\mathcal{R} \log \det H_{K,N} = \mathcal{R} \log \mathcal{A}_{N+1} - \mathcal{R} \log B_{N+1-q}(\chi, \chi). \quad (\text{MR19})$$

The omitted constant from MR18 has total coefficient zero, as in MR12.

There is also an explicit positive integral for the entire numerator. Put $t = d - r = n + m$ and, with each $S_j = c + iu_j$, define

$$\Phi_Z(S_1, \dots, S_t) = \sum_{|I|=r} z_I V_I \prod_{j=1}^t \chi_I(S_j), \quad \Delta_t(S) = \prod_{i < j} (S_j - S_i).$$

Then

$$\mathcal{A}_d = \frac{1}{t!} \int |\Delta_t(S) \Phi_Z(S_1, \dots, S_t)|^2 \prod_{j=1}^t d\mu(u_j). \quad (\text{MR20})$$

To prove this, expand the squared finite sum. For its (I, J) term, the determinant integration identity proved in [2] gives exactly $t! B_t(\chi_I, \chi_J)$ before the outside $1/t!$. All interchanges are finite and all moments exist. In particular the positive expression does not drop the interference terms inside the squared sum. It is not a probability density replacement for the source.

Mass is checked in these same formulas. Under the diagnostic change $d\mu \mapsto a d\mu$, $a > 0$, $\mathcal{A}_d$ scales as a^{d-r} and B_{d-q} scales as a^{d-q} . Their ratio scales as $a^{q-r} = a^m$, which is precisely the determinant scale of the rank- m original restricted Gram. Thus the actual mass in MR3 has not disappeared.

4 The exact Gamma receiver at the unresolved scale

JM21–28 [6] proves, on this simple-packet domain and in the same physical coordinate $S = c + iy$, a full-source comparison $\alpha_k \|P\|_\sigma^2 \leq \|P\|_\mu^2 \leq \beta_k \|P\|_\sigma^2$ for every degree at most $2q$, before any relation minimum. Here

$$\sigma(y) dy = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad L_k^{\text{form}} = \log(\beta_k/\alpha_k), \quad L_k^{\text{form}} = O_h(k \log^2(q+2)).$$

The positive constants include the exact convolution mass $c_k = (2\pi)^{k/2}/(\sqrt{2}\Gamma(k/2))$ before it cancels in their ratio. The complete comparison proof is retained in [6]; we do not replace it with an assertion about a pointwise density quotient. Applying the same affine minimum and then MR8 yields $\alpha_k H_{K,N}^\sigma \preceq H_{K,N} \preceq \beta_k H_{K,N}^\sigma$. Each log determinant difference lies between $m \log \alpha_k$ and $m \log \beta_k$. Pairing the four signs therefore proves again, with its exact finite constant,

$$|\mathcal{R} \log \det H_{K,N} - \mathcal{R} \log \det H_{K,N}^\sigma| \leq 2m L_k^{\text{form}} = o_h(kq). \quad (\text{MR21})$$

This is the earlier JM25 transfer, now with MR18–20 as its receiver.

For direct computation put $y_\alpha = (v_\alpha - c)/i$ and retain the original root-value columns: $P(v_\alpha) = P(c + iy_\alpha)$. Thus π and Z in these value coordinates are literally the same matrices. The monic Gamma polynomials and their full norms, with the source’s mass, are

$$p_0^\sigma = 1, \quad p_1^\sigma = y, \quad p_{j+1}^\sigma = yp_j^\sigma - j(j - \tfrac{1}{2})p_{j-1}^\sigma, \quad \gamma_j = \sqrt{2\pi} j! (\tfrac{1}{2})_j. \quad (\text{MR22})$$

These are the original Meixner–Pollaczek formulas used and derived in JM26 and its human sources [7, 6]. In S coordinates the monic polynomial is $i^j p_j^\sigma((S - c)/i)$, with the same norm. Hence the exact covariance and its invariant-image entry matrix are

$$C_d^\sigma = \sum_{j < d} \frac{f_j f_j^*}{\gamma_j}, \quad f_j = (p_j^\sigma(y_\alpha))_\alpha, \quad Z^T C_d^\sigma \bar{Z} = \sum_{j < d} \frac{(Z^T f_j)(Z^T f_j)^*}{\gamma_j}. \quad (\text{MR23})$$

The two phases i^j and $\bar{i}^j$ have multiplied to one in each kernel term; the original S multiplier and quotient coordinate maps remain as stated. MR11, MR18 and MR19 apply verbatim to this Gamma source, with their corresponding mixed B_t^σ . Together with MR21 this gives a fully specified finite expression whose four-return preserves the sought kq coefficient. That coefficient still requires evaluation; no individual current sign is assigned by these determinants.

5 Verification scope

The exact checker accompanying this note compares the full inverse restriction, its complementary determinant, the double mixed-minor sum, the positive integral identity in finite atomic test fixtures, and four-cutoff cancellation with nonreal frame entries and nonunit mass. Those fixtures test the proved finite algebra. They are not substituted for the original arithmetic measure or its actual period. The original period enters exactly through MR4–8; it has not been numerically sampled here. A numerical conditioning bound for the original large matrices is also not asserted.

References

- [1] G. Akemann and G. Vernizzi, *Characteristic Polynomials of Complex Random Matrix Models*, Nuclear Physics B 660 (2003), 532–556, <https://arxiv.org/abs/hep-th/0212051v2>, [https://doi.org/10.1016/S0550-3213\(03\)00221-9](https://doi.org/10.1016/S0550-3213(03)00221-9). Original author TeX read completely, including Th and deg.
- [2] Independent finite derivation for this programme, *The finite mixed and confluent Christoffel determinant on the original cyclic source*, 20 September 2026, accompanying complete TeX.
- [3] User-supplied programme continuation, *Source–boundary volume dynamics for the arithmetic control*, 12 September 2026, complete original NOTE.tex, (11)–(16).
- [4] Split-Zero programme, *An exact conductor frame and a fixed invariant-generator recursion*, 14 September 2026, OCF1–26, complete original source retained.
- [5] Split-Zero programme, *Full-window source products*, 19 September 2026, FW37–40 and FW46–52, complete original source retained.
- [6] Split-Zero programme, *The joint source in the original arithmetic metric: full spectrum, observation, and heat return*, 20 September 2026, result059, JM21–28 and independent KG1–21. This is the exact prior transfer used in MR21, not a new estimate attributed to Akemann–Vernizzi.

- [7] T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw; W. P. Reinhardt, *Orthogonal Polynomials*, NIST DLMF Chapter18, <https://dlmf.nist.gov/18.22.E8>, <https://dlmf.nist.gov/18.23.E7>; original Gamma norm calculation and mass retained in the complete JM/MT provider proofs.

Complete finite mixed and confluent determinant proof

6 Status, source, and exact receiving object

This note derives a finite identity from the equal-size case of G. Akemann and G. Vernizzi, *Characteristic Polynomials of Complex Random Matrix Models*, arXiv:hep-th/0212051v2, original author source `AkemannVernizzi.tex`, labels `Th`, `Ddef`, `proof1`, and `deg`. Its canonical disk-source identifier is PUBUNIT-688110C14C22CB41B5445BED. The original source's formula `deg` explicitly retains the factorial product for coincident arguments. We use divided derivatives throughout, so those factorials occur in the entries rather than as an outside multiplier.

The programme already proved the diagonal source/boundary determinant quotient in *Source-boundary volume dynamics for the arithmetic control*, supplied in `Tau_Toda_Volume_Control_2026-09-12.zip`, member `Tau_Toda_Volume_Control/NOTE.tex`, equations (11)–(16). The present diagonal formula is an exact kernel-jet re-expression of that quotient. The mixed formula retains two possibly different monic relations and their complex cross-products. No new general integrable system theorem, growing-degree estimate, or arithmetic-zero conclusion is asserted.

Fix the original finite full-order packet h , tensor order $k \geq 1$, and source

$$c = \frac{k}{2}, \quad S = c + iu, \quad v_h = \frac{2\xi}{h}, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad (1)$$

$$d\mu(u) = m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}.$$

Thus $\mu(\mathbb{R}) = \mu_h^k$, where $\mu_h = \int_{\mathbb{R}} w_h(t) dt$. This measure is never divided by its mass. The original polynomial source and arithmetic map are

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h, \quad J^{(k)}\mathcal{V}_{h,k} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P\left(\sum_i M_{s_i}\right)1. \quad (2)$$

Here $v_h = j_h(v_h)$ is the original invertible Taylor unit, $\chi = \chi_{h,k}$ is the actual monic cyclic relation, $q = \deg \chi$, $C = \mathbb{C}[S]/(\chi)$, and η is the original injection. The norm of this very source is

$$\langle P, Q \rangle = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) d\mu(u) = \langle \mathcal{V}_{h,k}P, \mathcal{V}_{h,k}Q \rangle. \quad (3)$$

The inner product is conjugate-linear in its first argument.

All polynomial moments of μ are finite, and every nonzero polynomial has positive norm. The following sufficient bound and its alternating-series/Stirling derivation are those of the original Toda note, Section 3.1, equations (7)–(8), retained here at their point of use. For completeness, its bound on the critical line is

$$w_h(t) \leq C_h(1 + |t|)^{11/2 - 2\deg h} e^{-\pi|t|/2} \quad \text{outside a compact interval.}$$

Indeed the alternating Dirichlet series has bounded partial sums, so summation by parts on $\Re s = 1/2$ gives an $O(1 + |s|)$ bound; the factor $1 - 2^{1-s}$ has modulus at least $\sqrt{2} - 1$; Stirling's vertical estimate, $s(s-1)$, and monic h give precisely the displayed sufficient bound. On the compact interval the full-order division makes v_h entire. It is not the zero function, and its zeros on the line form a discrete set. Thus $w_h > 0$ almost everywhere. For $0 < a < \pi/2$, Fubini and $|t_1 + \cdots + t_k| \leq \sum |t_i|$ give

$$\int e^{a|u|} d\mu(u) \leq \left(\int e^{a|t|} w_h(t) dt \right)^k < \infty.$$

The convolution is positive almost everywhere: for each further convolution the integrand is positive for almost every integration variable, since translations preserve null sets. The density is finite almost everywhere by its finite mass. A nonzero polynomial vanishes at only finitely many points of the line. Its squared norm is therefore strictly positive. These facts justify all finite integrations below.

7 Conventions and the finite identity

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = \{0\}$. For $d \geq 0$, let

$$e_d(S) = (1, S, \dots, S^{d-1})^T, \quad H_d = \int \overline{e_d(S)} e_d(S)^T d\mu, \quad D_d = \det H_d.$$

An empty vector and a 0×0 determinant have their usual meanings, and $D_0 = 1$. A coefficient column a represents $e_d(S)^T a$, whose squared norm is $a^* H_d a$. Positivity makes H_d invertible for $d > 0$. Gram–Schmidt in the original monomial order produces the unique monic orthogonal p_j ; let $\omega_j = \|p_j\|^2 > 0$. Its unit-triangular coefficient matrix has determinant one. Hence

$$D_d = \prod_{j=0}^{d-1} \omega_j, \quad K_d(v, \bar{w}) = \sum_{j=0}^{d-1} \frac{p_j(v) \overline{p_j(w)}}{\omega_j} = e_d(v)^T H_d^{-1} \overline{e_d(w)}. \quad (4)$$

For the final equality, the coefficient matrix A of the p_j satisfies $A^* H_d A = \text{diag}(\omega_j)$; solving for H_d^{-1} and multiplying the evaluation rows proves the formula. The kernel here is the source’s *bare* polynomial kernel, not a kernel with external density factors attached.

Let ψ be another monic polynomial of the same degree q . Factor the actual polynomials over $\mathbb{C}$, retaining all multiplicities:

$$\chi(S) = \prod_{a=1}^A (S - v_a)^{m_a}, \quad \psi(S) = \prod_{b=1}^B (S - w_b)^{\ell_b}, \quad \sum_a m_a = \sum_b \ell_b = q.$$

Roots are distinct within either displayed list; roots between the lists may coincide. Choose an order of each root list, and order each root’s derivative rows increasingly from zero. Define

$$V_\chi = \prod_{a < b} (v_b - v_a)^{m_a m_b}, \quad V_\psi = \prod_{a < b} (w_b - w_a)^{\ell_a \ell_b}. \quad (5)$$

For $n \geq 0$, put $d = n + q$ and define the $q \times q$ matrix

$$C_n(\chi, \psi)_{(a,r),(b,s)} = \frac{1}{r!s!} \partial_v^r \partial_{\bar{w}}^s K_d(v, \bar{w})|_{v=v_a, w=w_b}, \quad 0 \leq r < m_a, \quad 0 \leq s < \ell_b. \quad (6)$$

The conjugate variable is treated as an independent polynomial variable when differentiating. Thus its derivative is the complex conjugate of the corresponding ordinary derivative of p_j .

The requested mixed determinant, with its exact displayed orientation, is

$$B_n(\chi, \psi) = \det \left[\int S^i \chi(S) \overline{S^j \psi(S)} d\mu \right]_{i,j=0}^{n-1}, \quad B_0 = 1. \quad (7)$$

If $T_\chi, T_\psi : \mathbb{C}^n \rightarrow \mathbb{C}^d$ are the coefficient matrices of multiplication by their polynomials, the matrix displayed in (7) is $(T_\psi^* H_d T_\chi)^T$. In particular,

$$B_n(\chi, \psi) = \det(T_\psi^* H_d T_\chi), \quad B_n(\psi, \chi) = \overline{B_n(\chi, \psi)}. \quad (8)$$

The two cross-Gram orientations must not be identified entry by entry.

Theorem 1 (Finite mixed/confluent re-expression). *For the unchanged source (1) and all the integers and polynomials just specified,*

$$B_n(\chi, \psi) = D_{n+q} \frac{\det C_n(\chi, \psi)}{V_\chi \bar{V}_\psi}. \quad (9)$$

An empty root list has $V = 1$, and the empty kernel determinant is one. The mixed determinant can be complex or zero; no positivity of a mixed matrix is used. For $\psi = \chi$, its value is strictly positive when $n > 0$.

8 Finite determinant integration, with signs

Here and below $z_t = c + iu_t$ remains a point on the original vertical line, integrated with $d\mu(u_t)$. No area measure or matrix ensemble is introduced. For two ordered polynomial lists f_i, g_i , $0 \leq i < n$, expansion of the two determinants gives

$$\frac{1}{n!} \int \det[f_i(z_t)]_{t,i} \overline{\det[g_i(z_t)]_{t,i}} \prod_{t=1}^n d\mu(u_t) = \det \left[\int f_i(S) \overline{g_j(S)} d\mu \right]_{i,j}. \quad (10)$$

To verify the factor, a pair of permutations σ, τ contributes $\text{sgn}(\sigma) \text{sgn}(\tau) \prod_t M_{\sigma(t), \tau(t)}$, where $M_{ij} = \int f_i \bar{g}_j d\mu$. Reindexing by $i = \sigma(t)$ makes the second index $(\tau\sigma^{-1})(i)$. For each relative permutation there are exactly $n!$ choices of σ , giving $n! \det M$. All sums are finite and all products are absolutely integrable. For $n = 0$, both sides are one.

First assume all roots within each polynomial are simple and list them as $v_1, \dots, v_q$ and $w_1, \dots, w_q$. Use

$$\Delta_t(x_1, \dots, x_t) = \det[x_a^{j-1}]_{a,j=1}^t = \prod_{a < b} (x_b - x_a).$$

With the integrated rows first and the external rows last,

$$\Delta_d(z, v) = \Delta_n(z) \Delta_q(v) \prod_{t=1}^n \prod_{a=1}^q (v_a - z_t).$$

Thus

$$\det[z_t^i \chi(z_t)]_{t,i} = \Delta_n(z) \prod_t \chi(z_t) = (-1)^{nq} \frac{\Delta_d(z, v)}{\Delta_q(v)}.$$

The corresponding conjugate expression has the same real sign $(-1)^{nq}$. Their product has sign $(-1)^{2nq} = 1$. Consequently

$$B_n(\chi, \psi) = \frac{1}{n! \Delta_q(v) \overline{\Delta_q(w)}} \int \Delta_d(z, v) \overline{\Delta_d(z, w)} \prod_t d\mu(u_t). \quad (11)$$

Let $\phi_j = p_j / \sqrt{\omega_j}$, taking the positive square roots of the actual norms. Replacing monomials by their monic orthogonal polynomials has determinant one, so

$$\Delta_d(z, v) = D_d^{1/2} \det[\phi_j(x_a)]_{a=1, \dots, d; j=0, \dots, d-1}, \quad x = (z, v).$$

Expand this determinant along its last q rows. For a q -element column set J , in increasing order, the external minor is $\det \Phi_{v,J}$, and the integrated minor uses the complementary increasing column set

I . The Laplace sign is the sign determined by the sum of the selected row and column indices. Apply (10) to a pair of integrated minors. Orthonormality gives

$$\int \det \Phi_{z,I} \overline{\det \Phi_{z,I'}} \prod_t d\mu(u_t) = n! \det[\delta_{ij}]_{i \in I, j \in I'} = n! \mathbf{1}_{I=I'}.$$

When $I \neq I'$, the indicated square submatrix has a zero row; when they agree it is the identity. Hence only $J = J'$ survives, and its two Laplace signs multiply to one. The result is

$$\int \Delta_d(z, v) \overline{\Delta_d(z, w)} \prod_t d\mu(u_t) = n! D_d \sum_{|J|=q} \det \Phi_{v,J} \overline{\det \Phi_{w,J}} = n! D_d \det(\Phi_v \Phi_w^*).$$

The last equality follows by expanding the determinant of the product: terms with a repeated selected column cancel in pairs, and the remaining terms group into the displayed products of increasing-column minors. Its entries are $K_d(v_a, \bar{w}_b)$. Substituting in (11) proves (9) for simple roots. This proof also applies at $n = 0$, when there are no integrated rows.

9 Confluence and factorials

We prove the exact confluent determinant rule used here. Let a cluster of m ordered row arguments tend to v as $v + \varepsilon \alpha_t$, $t = 0, \dots, m-1$, with distinct constants α_t . For a polynomial row function F , Taylor expansion is the finite identity

$$F(v + \varepsilon \alpha_t) = \sum_{r \geq 0} \varepsilon^r \alpha_t^r \frac{F^{(r)}(v)}{r!}.$$

Multilinearity of the determinant kills every term with repeated derivative order within the cluster. The smallest possible total order is $0 + 1 + \dots + (m-1) = m(m-1)/2$. At that order, summing over the permutations of these orders produces exactly

$$\varepsilon^{m(m-1)/2} \det[\alpha_t^r]_{t,r=0}^{m-1} \det \left[\frac{F^{(r)}(v)}{r!} \right]_{r=0}^{m-1}$$

with the other rows left in their positions. There is no additional sign: both the rows in the scalar Vandermonde and the derivative rows have the increasing orders specified above. Higher-order terms carry strictly larger powers of ε .

Performing this operation in all clusters makes the within-cluster factors of the full Vandermonde exactly cancel the displayed scalar Vandermonde factors and powers of ε . Its between-cluster factors tend to $\prod_{a < b} (v_b - v_a)^{m_a m_b} = V_\chi$. Thus the limiting row determinant is the determinant of divided derivatives, divided by V_χ . Applying the identical argument to columns in the independent variable $\bar{w}$ gives the column factorials and $\overline{V_\psi}$. The roots within either cluster can be approached along the stated distinct paths; no real ordering or separation between the two polynomials is required.

The left side of (9) is a polynomial in the coefficients of χ and the conjugates of the coefficients of ψ , with finite moments as coefficients. It therefore tends to the exact repeated-root determinant. The right side tends, by the preceding calculation, to (6) divided by $V_\chi \overline{V_\psi}$. This completes the proof of Theorem 1.

If ordinary derivatives are used instead, define

$$F_\chi = \prod_a \prod_{r=0}^{m_a-1} r!, \quad F_\psi = \prod_b \prod_{s=0}^{\ell_b-1} s!.$$

The ordinary-derivative matrix has determinant $F_\chi F_\psi \det C_n$. Formula (9) then has the additional denominator $F_\chi F_\psi$. In the source's all-equal, unbarred example $\deg$, this is $\prod_{j=1}^{q-1} j!$. Its explicit factorial is consistent with the present calculation; the brief confluent prose is not being used to allege an error in the source.

10 Literal jet and remainder maps

Set $N = d - 1 = n + q - 1$. Define the divided-jet evaluation map

$$E_\chi : \mathcal{P}_N \longrightarrow \mathbb{C}^q, \quad P \longmapsto \left(\frac{P^{(r)}(v_a)}{r!} \right)_{(a,r)}. \quad (12)$$

The same symbol denotes its matrix in the original coefficient basis:

$$(E_\chi)_{(a,r),j} = \begin{cases} \binom{j}{r} v_a^{j-r} & j \geq r, \\ 0 & j < r. \end{cases}$$

It has kernel exactly $\chi \mathcal{P}_{n-1}$. Indeed Taylor's finite expansion at v_a shows that all its first m_a divided jets vanish if and only if $(S - v_a)^{m_a}$ divides P . Distinct linear factors are coprime, so their product divides P ; division decreases the degree by exactly q for a nonzero polynomial. The converse follows from the same expansion. At $n = 0$ this kernel is zero.

Let $r_\chi : \mathcal{P}_N \rightarrow \mathcal{P}_{q-1}$ be the unique remainder after monic division by χ , and let J_χ be the $q \times q$ matrix of E_χ restricted to $\mathcal{P}_{q-1}$. The preceding divisibility proof makes this restriction injective. Equal finite dimensions make it bijective, and

$$E_\chi = J_\chi r_\chi, \quad \mathbb{C}[S]/(\chi) \xrightarrow{[P] \mapsto (P^{(r)}(v_a)/r!)} \bigoplus_a \mathbb{C}[\epsilon_a]/(\epsilon_a^{m_a}) \quad (13)$$

is an algebra isomorphism, not only a vector-space identification. For multiplication, the coefficient of ϵ_a^r in the product of the two truncated Taylor series is $\sum_{t=0}^r P^{(t)}(v_a) Q^{(r-t)}(v_a)/(t!(r-t)!)$, which is $(PQ)^{(r)}(v_a)/r!$ by the Leibniz formula. This proves the algebra claim and fixes the target multiplication exactly.

Confluence applied to the ordinary evaluation matrix $[v_a^j]$ gives

$$\det J_\chi = V_\chi. \quad (14)$$

This also proves the exact orientation of its determinant. Permuting two root blocks of sizes m_a, m_b changes both sides by $(-1)^{m_a m_b}$; changing a root ordering has no effect on (9). Increasing derivative order within each block remains part of the definition.

Formula (4) and differentiation give the literal matrix identity

$$C_n(\chi, \psi) = E_\chi H_d^{-1} E_\psi^*. \quad (15)$$

If $k_{a,r}(S) = \sum_{j < d} p_j(S) \overline{p_j^{(r)}(v_a)/(r! \omega_j)}$, then $\langle k_{a,r}, P \rangle = P^{(r)}(v_a)/r!$. Consequently $C_n(\chi, \psi)$ is the standard, conjugate-first cross-Gram of these representing vectors, with the χ vector in its first slot and the ψ vector in its second slot. This is consistent with (8): the requested boundary matrix is transposed when expressed in coefficient cross-Gram notation, while the evaluation-kernel matrix is not.

11 The unchanged least-source-norm quotient

Now put $\psi = \chi$, and abbreviate $E = E_\chi$, $J = J_\chi$, $C_n = C_n(\chi, \chi)$. The surjectivity of E and positivity of H_d imply that C_n is positive definite for $q > 0$: for a nonzero column y , $E^*y \neq 0$, and $y^*C_n y = (E^*y)^*H_d^{-1}(E^*y) > 0$. For a prescribed jet column y , define

$$a_y = H_d^{-1}E^*C_n^{-1}y. \quad (16)$$

Then $Ea_y = y$. If $Eb = 0$, $a_y^*H_db = y^*C_n^{-1}Eb = 0$. Every other lift is $a_y + b$, and

$$\|a_y + b\|_{H_d}^2 = \|a_y\|_{H_d}^2 + \|b\|_{H_d}^2.$$

Thus (16) is the unique minimum-norm lift of the specified jet column; its norm is $y^*C_n^{-1}y$.

A remainder coefficient column $x \in \mathbb{C}^q$, in the exact ordered basis $1, S, \dots, S^{q-1}$, has jet column Jx . Its unique least-source-norm polynomial representative and quotient Gram are

$$\hat{R}_N = H_d^{-1}E^*C_n^{-1}J, \quad G_N = J^*C_n^{-1}J, \quad R_N = \mathcal{V}_{h,k}\hat{R}_N. \quad (17)$$

These are precisely the original quotient maps, because $r_\chi\hat{R}_N = J^{-1}E\hat{R}_N = I$, their images are orthogonal to the exact kernel $\chi\mathcal{P}_{n-1}$, and the preceding strict norm decomposition proves uniqueness. In particular the original arithmetic map is literally retained:

$$J^{(k)}R_N = \eta\pi_\chi\hat{R}_N = \eta.$$

No Taylor unit has been replaced by one, and no target jet norm has been substituted for the source norm. The jet norm C_n^{-1} was deduced from that source norm by minimization.

The determinant of (17), followed by Theorem 1, is

$$\boxed{\det G_N = \frac{|V_\chi|^2}{\det C_n(\chi, \chi)} = \frac{D_{n+q}}{B_n(\chi, \chi)}, \quad N = n + q - 1.} \quad (18)$$

For an independent comparison with the earlier Toda note, concatenate the coefficient columns of $1, S, \dots, S^{q-1}$ and $\chi, S\chi, \dots, S^{n-1}\chi$. Its leading-degree coefficient matrix is unit triangular, of determinant one. In this basis the source Gram has lower-right block $T_\chi^*H_dT_\chi$ of determinant $B_n(\chi, \chi)$. Subtracting from every initial lift its orthogonal projection to these boundary columns is a determinant-one column operation, with the matching conjugate row operation. It produces the orthogonal block Gram with diagonal blocks G_N and $T_\chi^*H_dT_\chi$. Therefore $D_d = \det G_N B_n(\chi, \chi)$. This is the original note's equation (16), now accompanied by its exact evaluation-jet factorization.

11.1 The norm of a monic polynomial in the original relation source

Equip the original input polynomials to M_χ with the pulled-back inner product

$$\langle P, Q \rangle_\chi = \langle \chi P, \chi Q \rangle = \int \overline{P(S)}Q(S)|\chi(S)|^2 d\mu.$$

This is the squared norm of the actual multiplication map into the same source, not a replacement of $d\mu$ by a probability measure. Its moments are finite and it is positive definite: multiplying a nonzero polynomial by the nonzero polynomial χ cannot yield zero, and the original norm is positive definite. Let p_j^χ be its monic orthogonal polynomials, with squared norms ν_j . The unit-triangular change from $1, S, \dots, S^{n-1}$ to $p_0^\chi, \dots, p_{n-1}^\chi$ gives

$$B_n(\chi, \chi) = \prod_{j=0}^{n-1} \nu_j.$$

Dividing this equality at consecutive orders, and then substituting (9), proves for every $n \geq 0$

$$\boxed{\nu_n = \frac{B_{n+1}(\chi, \chi)}{B_n(\chi, \chi)} = \omega_{n+q} \frac{\det C_{n+1}(\chi, \chi)}{\det C_n(\chi, \chi)}}. \quad (19)$$

The factors $|V_\chi|^2$ cancel because the relation and its exact multiplicities are unchanged at consecutive orders. The denominator is positive; for $q = 0$ both kernel determinants are empty and $\nu_n = \omega_n$. Under a scalar comparison $d\mu \mapsto \lambda d\mu$, the kernel-determinant ratio is unchanged and ω_{n+q} scales by λ , so the full original mass is retained in ν_n .

Akemann–Vernizzi’s source discusses the general-size expression at **Lnorm** as expected and checked in several small sizes; the equal-degree determinant theorem above plus this finite Gram argument supplies its norm expression for the present original relation source, including repeated roots. This is not a priority claim about the later literature, and it does not claim that every separate orthogonal-polynomial formula discussed there has been proved here. The original Toda note already identifies $\nu_n = B_{n+1}/B_n$ in its equation (19).

12 Boundary cases

If $n = 0$ and $q > 0$, then $d = q$, $E_\chi = J_\chi$, and

$$C_0(\chi, \psi) = J_\chi H_q^{-1} J_\psi^*, \quad \det C_0 = \frac{V_\chi \overline{V_\psi}}{D_q}.$$

Thus (9) reads $1 = 1$, not an unstated limiting convention. There are no admitted boundary columns, the quotient section is the identity in coefficients, and $G_{q-1} = H_q$.

If $q = 0$, both monic degree-zero polynomials are 1, their root lists and jets are empty, and $B_n(1, 1) = D_n$. Both sides of (9) agree. The quotient $\mathbb{C}[S]/(1)$ is the zero module, the jet target is the zero vector space, and its determinant is one. Its least-norm representative of its sole class is zero; all of $\mathcal{P}_{n-1}$ is the relation subspace. No positive-dimensional inverse is asserted. For $n = q = 0$, take $\mathcal{P}_{-1} = 0$; every determinant in the statement equals one. One may read all 0×0 maps as the unique maps on the zero space, but no inverse formula is needed to prove this case.

13 Mass scaling without changing the source

To expose every mass power, consider only as an algebraic comparison $d\mu \mapsto \lambda d\mu$, with $\lambda > 0$. Monic orthogonal polynomials stay the same, their norms become $\lambda \omega_j$, and

$$H_d \mapsto \lambda H_d, \quad D_d \mapsto \lambda^d D_d, \quad K_d \mapsto \lambda^{-1} K_d, \quad C_n \mapsto \lambda^{-1} C_n, \quad B_n \mapsto \lambda^n B_n.$$

The right side of (9) scales as $\lambda^{d-q} = \lambda^n$, exactly as its left side. The quotient section (17) stays unchanged and $G_N \mapsto \lambda G_N$, so its determinant scales as λ^q . These identities show where the original total mass μ_h^k is retained. No such scaling is actually performed on (1).

14 The exact $S = c + iu$ coordinate maps and phases

The coefficient change from S to u is the upper triangular matrix

$$(L_d)_{rj} = \begin{cases} \binom{j}{r} c^{j-r} i^r & r \leq j, \\ 0 & r > j, \end{cases} \quad 0 \leq r, j < d, \quad \det L_d = i^{d(d-1)/2}.$$

It is defined by the exact equality $P(c + iu) = e_d(u)^T L_d a$ for $P(S) = e_d(S)^T a$. If H_d^u is the original u -monomial Gram with the same $d\mu(u)$, then

$$H_d = L_d^* H_d^u L_d, \quad D_d = \det H_d^u. \quad (20)$$

The determinant equality follows only after retaining and multiplying the phases $\overline{\det L_d} \det L_d = 1$.

Let Q_j be the monic u -orthogonal polynomial. The polynomial $i^j Q_j((S - c)/i)$ is monic in S , has the required orthogonality, and has norm $\|Q_j\|^2$; uniqueness gives

$$p_j(S) = i^j Q_j((S - c)/i), \quad \omega_j = \|Q_j\|^2.$$

For $\zeta = (v - c)/i$, $\eta = (w - c)/i$, each summand of the kernel has the paired phases $i^j \bar{i}^j = 1$. Hence

$$K_d(v, \bar{w}) = K_d^u(\zeta, \bar{\eta}), \quad \partial_v = i^{-1} \partial_\zeta, \quad \partial_{\bar{w}} = \bar{i}^{-1} \partial_{\bar{\eta}} = i \partial_{\bar{\eta}}.$$

Define the chart polynomials $\tilde{\chi}(u) = i^{-q} \chi(c + iu)$ and $\tilde{\psi}(u) = i^{-q} \psi(c + iu)$. They are monic; the actual source multipliers remain $\chi(c + iu) = i^q \tilde{\chi}(u)$ and $\psi(c + iu) = i^q \tilde{\psi}(u)$. For their roots $\zeta_a = (v_a - c)/i$ and $\eta_b = (w_b - c)/i$, put

$$R_\chi = \sum_a \frac{m_a(m_a - 1)}{2}, \quad P_\chi = \sum_{a < b} m_a m_b, \quad R_\chi + P_\chi = \frac{q(q - 1)}{2},$$

and similarly for ψ . Let Λ_χ be the diagonal jet matrix with entries i^{-r} in row (a, r) . The chain rule gives exact matrix identities

$$E_\chi^S = \Lambda_\chi E_{\tilde{\chi}}^u L_d, \quad J_\chi^S = \Lambda_\chi J_{\tilde{\chi}}^u L_q, \quad C_n^S(\chi, \psi) = \Lambda_\chi C_n^u(\tilde{\chi}, \tilde{\psi}) \Lambda_\psi^*. \quad (21)$$

Their determinant phases are

$$\det C_n^S = i^{-R_\chi + R_\psi} \det C_n^u, \quad V_\chi^S = i^{P_\chi} V_{\tilde{\chi}}^u, \quad \overline{V_\psi^S} = i^{-P_\psi} \overline{V_{\tilde{\psi}}^u}.$$

The resulting phase of the ratio in (9) is $i^{-R_\chi + R_\psi - P_\chi + P_\psi} = 1$, because both polynomials have the same degree q . This checks the mixed orientation even when their multiplicity patterns differ.

For the boundary determinants, changing the n input monomials introduces L_n and its adjoint; the two degree- q multiplier phases contribute $\bar{i}^q i^q = 1$ to the coefficient cross-Gram. Thus

$$B_n^S(\chi, \psi) = B_n^u(\tilde{\chi}, \tilde{\psi})$$

with the measure still $d\mu$. The quotient map is the literal algebra isomorphism

$$\mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[u]/(\tilde{\chi}), \quad [P(S)] \longmapsto [P(c + iu)].$$

It is well defined since the relation becomes $i^q \tilde{\chi}$; its inverse substitutes $u = (S - c)/i$. It preserves source norms by (3). On remainder coefficient columns it is L_q , so $G_N^S = L_q^* G_N^u L_q$, again with determinant multiplier one. This is a proved coordinate transport, not a new relation or a new measure.

15 What has and has not been checked

The companion `check_identities.py` uses exact rational complex arithmetic to verify determinant integration, mixed orientations, confluent derivatives, nonunit mass powers, all boundary cases, and the minimum-section equations. Its finite atomic fixtures test the polynomial and matrix identities proved above; they are not substituted for the programme density, and they are not a numerical computation of the original arithmetic moments. The proof applies to those original moments by Sections 6–8. No probabilistic ensemble interpretation is required at any stage.

Independent verification and direct Gamma norm derivation

16 Independent review of the actual kernel receiver

The root derivation `../ORIGINAL_KERNEL_MIXED_DETERMINANT.tex`, MR1–23, was read completely. No sign, transpose, factorial, degree-shift, or source-mass error was found in its finite formulas. This is an independent mathematical review, not a publication receipt or an evaluation of the original period coefficients.

16.1 Original coefficient receiver

The complete original `ORIGINAL_CONDUCTOR_STRIP_FRAME.tex`, OCF1–26, was read at:

`ORIGINAL_CONDUCTOR_STRIP_FRAME.tex`.

OCF7 proves $N_k S_k = I$ and $\ker N_k = \text{im } M_A$. OCF17 proves that the conductor image is contained in the actual invariant image, with its original factor 5 in the Reynolds lift. OCF19 gives $\text{rank } Z_k = 8k - 32$ for the stated $k \geq 9$ domain. OCF21–22 prove the complete coefficient quotient and its right inverse. Therefore, with the root derivation's exact columns $Z = [M_A, S_k \ Z_k]$, applying N_k to a column relation proves first that the Z_k coefficients vanish and then that the M_A coefficients vanish. Hence Z has full column rank and its image is exactly the invariant image, not merely the conductor image. The count is

$$r = (k - 7)^2 + (8k - 32) = k^2 - 6k + 17, \quad q = (k + 1)^2, \quad m = q - r = 8k - 16.$$

OCF24 identifies the biform coefficient pairing with the original complex-linear primary dual pairing in the same lexicographic order. Thus a coefficient column z acts on a primary value column y as $z^T y$. It follows that the actual kernel's value-space image is $\ker Z^T = \text{im } \pi^T$, because $\pi Z = 0$ and both spaces have dimension m . No conjugation belongs in this coefficient dualization.

16.2 Complementary determinant and its frame constants

Set $U = \pi^T$ and $V = \overline{Z}$. Then

$$U^* V = \overline{\pi Z} = 0, \quad U^* U = \overline{\pi \pi^T}, \quad V^* V = Z^T \overline{Z}.$$

Both Gram matrices are positive definite. Using their positive square roots only as a finite proof device makes $U_0 = U(U^* U)^{-1/2}$ and $V_0 = V(V^* V)^{-1/2}$ into complementary orthonormal frames. For any positive definite complete covariance C , the matrix in that orthonormal frame is

$$\begin{pmatrix} A & B \\ B^* & D \end{pmatrix}.$$

Block elimination gives $\det C = \det D \det(A - B D^{-1} B^*)$, while the upper inverse block is $(A - B D^{-1} B^*)^{-1}$. Restoring both original frame factors gives exactly

$$\det(U^* C^{-1} U) = \frac{\det(U^* U)}{\det(V^* V)} \frac{\det(V^* C V)}{\det C} = \frac{\det(\overline{\pi \pi^T})}{\det(Z^T \overline{Z})} \frac{\det(Z^T C \overline{Z})}{\det C}.$$

This verifies MR11. Replacing $\overline{Z}$ by Z , or deleting either frame factor, is not justified. The root derivation makes neither replacement.

16.3 Mixed expansion, orientation and integral

For increasing root subsets I, J of cardinality r , twice applying finite Cauchy–Binet gives

$$\det(Z^T C_d \overline{Z}) = \sum_{I, J} \det Z_I \overline{\det Z_J} \det(C_d)_{I, J}.$$

The external kernel has $d=N+1$ terms. The mixed finite formula therefore applies with relation degree r and integrated determinant size $t=d-r$, yielding

$$\det(C_d)_{I, J} = \frac{V_I \overline{V_J}}{D_d} B_{d-r}(\chi_I, \chi_J).$$

The root and minor orders agree, so no extra permutation sign occurs. Since $n=d-q$ and $m=q-r$, the numerator size is exactly $t=n+m$. The denominator's full relation size is n . At the first admitted cutoff $N=q-1$, this is numerator size m and denominator $B_0=1$, not a negative degree.

For $a_I = z_I V_I$, let $\Phi = \sum_I a_I \prod_{j=1}^t \chi_I(S_j)$. Expanding $|\Delta_t(S)\Phi|^2$ gives coefficients $a_I \bar{a}_J$; determinant integration gives exactly $t! B_t(\chi_I, \chi_J)$ for each term. Thus

$$\mathcal{A}_d = \frac{1}{t!} \int |\Delta_t(S)\Phi(S_1, \dots, S_t)|^2 \prod_j d\mu(u_j).$$

This verifies MR20 including interference terms and the factorial. It is the norm of the complete finite sum. It is not the sum obtained by dropping the off-diagonal terms.

The full-root denominator identity is $\det C_d = |V_\chi|^2 B_n(\chi, \chi)/D_d$. Substitution gives MR18. Under the diagnostic comparison $d\mu \rightarrow a \, d\mu$, A_d scales as a^{d-r} and B_n as a^{d-q} ; their ratio scales as a^m , exactly the rank- m restricted metric determinant. The coefficient frame factors are fixed across all four cutoffs, so their logarithms cancel with coefficients $1+1-1-1$. This proves MR19 without assuming monotonicity of the individual errors or of a mixed summand.

16.4 Gamma receiver and its full mass

Actual original sources read for this part:

- `JOINT_MINIMUM_RECEIVERS.tex`, lines 276–414, containing the full JM21–28 section.
- Complete `KERNEL_GAMMA_INDEPENDENT.md`, KG1–21.
- `COERCIVITY_INDEPENDENT.md`, lines 90–132, containing C1–C4.

All three are in `joint_minimum_arrival`. The prior coercivity theorem used to obtain the source comparison was not replayed here. Its proved comparison is the explicit imported input, as in the root note; the root note does not claim to derive that estimate from Akemann–Vernizzi.

Here is a separate derivation of the exact recurrence and norms in MR22. The beta-integral Fourier calculation in KG1–2 gives

$$\int e^{-ity} \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy = \sqrt{2\pi} (\cosh t)^{-1/2}.$$

The original Gamma tail estimate gives an integrable exponential majorant on closed subintervals of $|\operatorname{Re} s| < \pi/2$. The moment-generating integral is therefore analytic there, and continuation of the displayed Fourier identity gives, for real $|s| < \pi/2$,

$$\int e^{sy} d\sigma(y) = \sqrt{2\pi} (\cos s)^{-1/2}.$$

For real z, w sufficiently close to zero define

$$F(z, y) = (1 + z^2)^{-1/4} e^{y \arctan z} = \sum_{j \geq 0} p_j(y) z^j / j!.$$

Taylor coefficients in z are polynomials in y of degree j , and their top coefficient is one. The differential identity

$$(1 + z^2) \partial_z F = (y - z/2) F$$

gives $p_0=1$, $p_1=y$, and $p_{j+1}=y p_j - j(j-1/2)p_{j-1}$ after coefficient comparison. The same exponential majorant justifies differentiating the following integral in both real parameters to every finite order:

$$\begin{aligned} \int F(z, y) F(w, y) d\sigma(y) &= \sqrt{2\pi} [(1 + z^2)(1 + w^2)]^{-1/4} [\cos(\arctan z + \arctan w)]^{-1/2} \\ &= \sqrt{2\pi} (1 - zw)^{-1/2}. \end{aligned}$$

The last identity uses the positive real branches near zero and $\cos(\arctan z + \arctan w) = (1 - zw) / \sqrt{(1 + z^2)(1 + w^2)}$. Comparing the coefficient of $z^j w^1$ proves

$$\int p_j(y) p_l(y) d\sigma(y) = \delta_{jl} \sqrt{2\pi} j! (1/2)_j.$$

The polynomials have real coefficients, so this is their conjugate-first orthogonality on the real line. It proves MR22 with mass $\gamma_0 = \sqrt{(2\pi)}$, without omitting the scale $dy=2dx$ implicit in the original Gamma density.

The original S -monic polynomial is $i^j p_j((S-c)/i)$; its squared norm is unchanged. Its root evaluation column is $i^j f_j$. Consequently the covariance summand is $i^j f_j (i^j f_j f_j)^* / \gamma_j = f_j f_j^* / \gamma_j$. The value of every quotient polynomial at every root remains the same under the literal coordinate substitution, so the bilinear coefficient frames π, Z do not change in value coordinates. This proves MR23 and its compatibility with MR8–11.

Finally the full-source comparison through degree $2q$ applies to every candidate in each identical remainder fibre. Taking the minima, and only then restricting by the fixed actual kernel frame, gives the two matrix inequalities of MR21. Each log determinant error lies in the common interval $[m \log \alpha_k, m \log \beta_k]$. Two paired differences give the exact factor $2m \log(\beta_k / \alpha_k)$. With $m=8k-16$, $q=(k+1)^2$, and the imported width $O_h(k \log^2(q+2))$, division by kq gives a quantity tending to zero. This checks the stated Gamma transfer without evaluating the unresolved determinant coefficient.

16.5 Exact finite checks

`check_receiver.py` independently verified 43 identities in normal and optimized Python. Its three nonreal coefficient frames include $(q, m, r) = (2, 1, 1), (3, 2, 1), (3, 1, 2)$. The checks compare the inverse restriction, complementary determinant, full mixed sum, positive squared-sum integral, first admitted cutoff, mass exponent, and exact four-cutoff ratio. `RECEIVER_CHECKS.json` retains the successful optimized output; the normal run returned the same 43 passing checks. The fixtures retain source mass 21 but do not sample the arithmetic period or replace the original source.

The complete original conductor and invariant coefficient frame

17 An exact conductor frame and a fixed invariant-generator recursion

Retain the original simple-quartet data, $m = 1$, $k = 4l + 1$, $l \geq 1$, the original root order $(++)$, $(+-)$, $(-+)$, $(--)$, and the complete original period and amplitude matrix of OQC1:

$$\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}.$$

Thus $\det \mathbf{H} \neq 0$ and $Q(x) = Q_0(\mathbf{H}^{-1}x)$, where $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. Use $gf(x) = f(D^{-1}x)$, $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and the original invariant algebra $\mathcal{S} = \mathbb{C}[x_1, x_2, x_3, x_4]^{(g)}$. This calculation applies on the actual five-orbit period locus, which includes every original logarithm branch with $|u| \geq R_*$ by PQO and RPC. On that locus the five prime polynomials $Q_a = g^a Q$, $0 \leq a < 5$, are pairwise nonassociate. Keep their actual products

$$\mathcal{A} = \prod_{a=1}^4 Q_a, \quad \mathcal{H}_Q = Q\mathcal{A} = \prod_{a=0}^4 Q_a, \quad \deg \mathcal{A} = 8, \quad \deg \mathcal{H}_Q = 10. \quad (\text{OCF1})$$

No coefficient in these polynomials is replaced. The exact orbit decision and the period domain are those of the cited original calculations. The construction below treats every possible nonzero leading coefficient of their actual restriction.

17.1 The original Segre coefficient isomorphism

For $d \geq 0$ let $\mathcal{B}_d$ be the space of biforms of bidegree (d, d) in (z_0, z_1) and (w_0, w_1) . Its ordered basis is

$$e_{ab}^{(d)} = z_0^a z_1^{d-a} w_0^b w_1^{d-b}, \quad 0 \leq a, b \leq d. \quad (\text{OCF2})$$

Pairs are ordered lexicographically, first by a , then by b . The order is used only for the exact coefficient elimination. Set $\mathcal{B}_d = 0$ for $d < 0$. Multiplication satisfies $e_{ab}^{(d)} e_{ij}^{(e)} = e_{a+i, b+j}^{(d+e)}$. The original substitution is

$$t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T, \quad x = \mathbf{H}t, \quad (\text{OCF3})$$

$$\mathcal{P}_d = \mathbb{C}[x_1, x_2, x_3, x_4]_d, \quad \psi_d : \mathcal{P}_d \longrightarrow \mathcal{B}_d, \quad f \longmapsto f(\mathbf{H}t).$$

The induced map $R_d = \mathcal{P}_d / (Q\mathcal{P}_{d-2}) \longrightarrow \mathcal{B}_d$ is an isomorphism. Here is a coefficient proof with the given orientation. For each (a, b) , choose an integer h between $\max(0, a + b - d)$ and $\min(a, b)$. The monomial $t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{d-a-b+h}$ maps to $e_{ab}^{(d)}$, proving surjectivity. Two monomials with the same pair differ in their exponent vectors by an integer multiple of $(1, -1, -1, 1)$. After removing their common monomial, their difference is divisible by $t_{++}t_{--} - t_{+-}t_{-+}$, using $A^n - B^n = (A - B) \sum_{j=0}^{n-1} A^{n-1-j} B^j$. A polynomial in the kernel has coefficient sum zero within every pair, so it is a sum of such differences. The reverse inclusion in the kernel follows by direct substitution. The invertible substitution $x = \mathbf{H}t$ then gives exactly the asserted kernel. Thus these maps preserve multiplication as well as every grading.

Write the full conductor restriction as

$$A = \psi_8(\mathcal{A}) = \sum_{r=0}^8 \sum_{s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0. \quad (\text{OCF4})$$

To verify its nonvanishing, Q is a rank-four irreducible quadric, hence prime in the polynomial ring, and none of $Q_1, \dots, Q_4$ is its associate. Therefore Q does not divide their product. The kernel calculation OCF3 proves OCF4. The ring $\mathcal{B} = \bigoplus_{d \geq 0} \mathcal{B}_d$, with the indicated diagonal grading, embeds in the four-variable polynomial ring; it is an integral domain. Multiplication by A is consequently injective on every $\mathcal{B}_d$.

17.2 Every leading-coefficient case gives an exact strip frame

Let (r_*, s_*) be the lexicographically largest pair for which $a_{r_* s_*} \neq 0$, and put $a_* = a_{r_* s_*}$. This is an actual nonzero coefficient of OCF4. For $d \geq 8$ define

$$\begin{aligned} P_d &= \{(r_* + i, s_* + j) : 0 \leq i, j \leq d - 8\}, \\ E_d^{\text{strip}} &= \{0, \dots, d\}^2 \setminus P_d, \quad W_d = \mathbb{C}^{E_d^{\text{strip}}}. \end{aligned} \tag{OCF5}$$

For $0 \leq d < 8$ put $P_d = \emptyset$, $E_d^{\text{strip}} = \{0, \dots, d\}^2$ and use the same definition of W_d ; for $d < 0$ put $W_d = 0$. Let $S_d : W_d \hookrightarrow \mathbb{C}^{(d+1)^2}$ insert zeros in the positions P_d . All coefficient columns below use OCF2.

The following finite elimination defines $N_d : \mathbb{C}^{(d+1)^2} \rightarrow W_d$ and, for $d \geq 8$, $V_d : \mathbb{C}^{(d+1)^2} \rightarrow \mathbb{C}^{(d-7)^2}$. Starting with the complete input biform f , traverse all $(i, j) \in \{0, \dots, d - 8\}^2$ in decreasing lexicographic order. At the step (i, j) , let c be the current coefficient at $(r_* + i, s_* + j)$, record $b_{ij} = c/a_*$, and replace the current biform by

$$f_{\text{current}} - b_{ij} A e_{ij}^{(d-8)}. \tag{OCF6}$$

At completion take the coefficients on E_d^{strip} as $N_d f$ and take (b_{ij}) as $V_d f$. If $d < 8$, N_d is the identity and V_d is the zero map into the zero space. The exact multiplication matrix $M_{A,d}$ gives

$$S_d N_d + M_{A,d} V_d = I, \quad N_d S_d = I_{W_d}, \quad \ker N_d = A \mathcal{B}_{d-8}. \tag{OCF7}$$

To prove these identities, the largest term of $A e_{ij}^{(d-8)}$ is $a_* e_{r_*+i, s_*+j}^{(d)}$; every other term is strictly smaller in the same lexicographic order. Therefore each step zeros its designated position without altering an earlier zero. Summing all subtractions gives the first identity. Input supported on the strip produces zero coefficients at every elimination step, giving the second. Finally, the leading term of Ab , for a nonzero homogeneous biform b , is the product of the two leading terms and belongs to P_d . Indeed every other product is smaller in the additive lexicographic order. Thus $A \mathcal{B}_{d-8} \cap S_d W_d = 0$. The first identity now proves the kernel formula, the uniqueness of the decomposition, and all three identities in OCF7. All operations are linear in the input; their coefficients are explicit rational expressions in the original a_{rs} with powers of the displayed a_* as denominators. The polynomial A itself remains the full polynomial OCF4.

In particular the quotient map induced by N_d is the exact isomorphism

$$\mathcal{B}_d / (A \mathcal{B}_{d-8}) \xrightarrow{\sim} W_d, \quad n_d := \dim W_d = \begin{cases} (d+1)^2, & 0 \leq d < 8, \\ 16d - 48, & d \geq 8. \end{cases} \tag{OCF8}$$

The second formula is $(d+1)^2 - (d-7)^2 = 16d - 48$. The possible choices $(r_*, s_*) \in \{0, \dots, 8\}^2$ form an exhaustive finite partition: the chart for a pair specifies that its coefficient is nonzero and every larger coefficient is zero. Hence all actual coefficient cases have been included. At the first original degree $k = 5$ this construction retains all 36 coordinates; at every original $k \geq 9$ it retains $16k - 48$ coordinates.

17.3 Twenty-five fixed invariant monomials generate all degrees

Define the following fixed finite sets of exponent vectors:

$$G_e = \{\nu \in \mathbb{N}^4 : |\nu| = e, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}, \quad 1 \leq e \leq 5, \quad G = \bigcup_{e=1}^5 G_e. \quad (\text{OCF9})$$

Here G_1 is empty. The complete remaining lists, in the original x_1, x_2, x_3, x_4 order, are

e	G_e	
2	$(0, 1, 1, 0), (1, 0, 0, 1)$	
3	$(0, 0, 2, 1), (0, 1, 0, 2), (1, 2, 0, 0), (2, 0, 1, 0)$	
4	$(0, 0, 1, 3), (0, 2, 2, 0), (0, 3, 0, 1), (1, 0, 3, 0),$ $(1, 1, 1, 1), (2, 0, 0, 2), (3, 1, 0, 0)$	(OCF10)
5	$(0, 0, 0, 5), (0, 0, 5, 0), (0, 1, 3, 1), (0, 2, 1, 2),$ $(0, 5, 0, 0), (1, 0, 2, 2), (1, 1, 0, 3), (1, 3, 1, 0),$ $(2, 1, 2, 0), (2, 2, 0, 1), (3, 0, 1, 1), (5, 0, 0, 0).$	

Thus $(|G_1|, \dots, |G_5|) = (0, 2, 4, 7, 12)$ and $|G| = 25$. These lists can also be obtained directly by choosing $0 \leq \nu_1, \nu_2, \nu_3 \leq e$ with their sum at most e , putting $\nu_4 = e - \nu_1 - \nu_2 - \nu_3$, and applying the single displayed congruence, which verifies completeness of each list.

Every positive-degree invariant monomial is a product of members x^ν , $\nu \in G$. To prove this, list the weights of its individual factors in any order. If its degree is at most five, the entire list gives a member of G . If its degree exceeds five, consider the six partial sums of the first five weights, including the empty partial sum, in $\mathbb{Z}/5\mathbb{Z}$. Two coincide; the nonempty block between them has length at most five and total weight zero. Its factors give a divisor x^ν with $\nu \in G$. The complementary monomial again has total weight zero and smaller degree. Induction completes the factorization. Because the action is diagonal, a polynomial is invariant exactly when all its nonzero monomials have weight zero: comparison of each monomial coefficient in $gf = f$ proves this statement. Thus

$$\mathcal{J} = \mathbb{C}[x^\nu : \nu \in G]. \quad (\text{OCF11})$$

All intermediate degrees here are degrees in the original coefficient algebra. This construction introduces no new source distribution and leaves the original source orders 1 and k intact.

For each $\nu \in G_e$ the complete generator biform is

$$f_\nu = \psi_e(x^\nu) = \prod_{j=1}^4 (H_{j1}z_0w_0 + H_{j2}z_0w_1 + H_{j3}z_1w_0 + H_{j4}z_1w_1)^{\nu_j} = \sum_{a,b=0}^e c_{\nu,ab} e_{ab}^{(e)}. \quad (\text{OCF12})$$

For an entirely explicit coefficient formula, sum over all nonnegative 4 by 4 integer arrays (n_{jt}) satisfying $\sum_t n_{jt} = \nu_j$, $\sum_j (n_{j1} + n_{j2}) = a$ and $\sum_j (n_{j1} + n_{j3}) = b$. Then

$$c_{\nu,ab} = \sum_{(n_{jt})} \prod_{j=1}^4 \left(\frac{\nu_j!}{\prod_{t=1}^4 n_{jt}!} \prod_{t=1}^4 H_{jt}^{n_{jt}} \right). \quad (\text{OCF13})$$

The multinomial theorem proves the formula, including every original period and amplitude entry. Every factor has degree at most five and at most 36 possible biform coefficients.

17.4 Exact multiplication on the strip and the invariant recursion

Let $\mathcal{L} = \mathcal{B}/(AB)$, a graded algebra. For $\nu \in G_e$ and $d \geq e$ define

$$m_{\nu,d} : W_{d-e} \longrightarrow W_d, \quad m_{\nu,d} = N_d M_{f_\nu} S_{d-e}. \quad (\text{OCF14})$$

Here M_{f_ν} is the exact coefficient convolution:

$$(M_{f_\nu} S_{d-e} v)_{ab} = \sum_{i,j} c_{\nu,a-i,b-j} (S_{d-e} v)_{ij},$$

with out-of-range indices giving zero. Therefore every entry of OCF14 is supplied by the finite formulas OCF6 and OCF13. For a representative $h \in \mathcal{B}_{d-e}$, OCF7 writes $h - S_{d-e} N_{d-e} h = Ab$. Multiplication by f_ν preserves this ideal. Applying N_d yields the exact identity

$$N_d(f_\nu h) = m_{\nu,d} N_{d-e} h. \quad (\text{OCF15})$$

This proves that OCF14 is multiplication by the specified element of $\mathcal{L}$, including its independence of representatives.

The original invariant image is $J_d = \psi_d(\mathcal{J}_d) \subseteq \mathcal{B}_d$. Set $\bar{J}_d = N_d J_d \subseteq W_d$, and set $\bar{J}_d = 0$ for $d < 0$. The exact recursion is

$$\bar{J}_0 = \mathbb{C} \cdot 1, \quad \bar{J}_d = \sum_{\substack{\nu \in G_e \\ 2 \leq e \leq 5, e \leq d}} m_{\nu,d} \bar{J}_{d-e} \quad (d \geq 1). \quad (\text{OCF16})$$

Indeed OCF11 factors every invariant monomial of degree $d > 0$ as x^ν times an invariant monomial of degree $d - e$. Applying ψ , then OCF15, gives the inclusion from left to right. Conversely every summand is the image of a product of two invariant polynomials, so it belongs to the left side. Linearity proves equality for the complete invariant space. In particular the recursion gives $\bar{J}_1 = 0$ and includes all small degrees.

The conductor ideal is contained in this invariant image with its original trace coefficient:

$$AB_{d-8} \subseteq J_d, \quad [5R(\mathcal{A}h)]_{(Q)} = [\mathcal{A}h]_{(Q)}, \quad R = \frac{1}{5} \sum_{a=0}^4 g^a. \quad (\text{OCF17})$$

To prove the identity, $g^a \mathcal{A}$ is the product of all five orbit equations except Q_a ; for $a \neq 0$ it contains Q . Thus all four corresponding terms vanish modulo Q , and the $a = 0$ term is the displayed product. Every biform in $\mathcal{B}_{d-8}$ has a polynomial lift by OCF3, so the identity proves the inclusion. It also shows exactly why the quotient used in OCF16 retains the original invariant image.

17.5 Exact ranks, a finite pivot atlas, and the original quotient map

For $d \geq 0$ put

$$d_0(d) = \frac{\binom{d+3}{3} + 4\epsilon_d}{5}, \quad \epsilon_d = \mathbf{1}_{d \equiv 0 \pmod{5}} - \mathbf{1}_{d \equiv 1 \pmod{5}}, \quad d_0(d) = 0 \quad (d < 0). \quad (\text{OCF18})$$

This is the number of degree- d weight-zero monomials: the identity character has generating function $(1 - z)^{-4}$, each of the four nonidentity characters has generating function $\prod_{j=1}^4 (1 - z\zeta^j)^{-1} = (1 - z)/(1 - z^5)$, and averaging their coefficients gives OCF18. Every invariant polynomial divisible by Q is divisible by all five Q_a , hence by $\mathcal{H}_Q$, since they are pairwise nonassociate primes. Its quotient by this product is invariant: apply g and cancel the nonzero invariant product in the integral domain.

Conversely each $\mathcal{H}_Q$ times an invariant polynomial is invariant and vanishes modulo Q . Consequently $\ker(\mathcal{J}_d \rightarrow J_d) = \mathcal{H}_Q \mathcal{J}_{d-10}$, and injection of multiplication by the nonzero product gives $\dim J_d = d_0(d) - d_0(d-10)$. Injection of multiplication by A , OCF7 and OCF17 then prove

$$j_d := \dim \bar{J}_d = d_0(d) - d_0(d-10) - \max(d-7, 0)^2. \quad (\text{OCF19})$$

In particular $j_0 = 1$, $j_1 = 0$, $j_5 = 12$, and $j_d = 8d - 32$ for $d \geq 8$. For the last formula at $d \geq 10$, $\epsilon_d = \epsilon_{d-10}$ and direct expansion gives $\binom{d+3}{3} - \binom{d-7}{3} = 5(d^2 - 6d + 17)$; subtracting $(d-7)^2$ gives $8d - 32$. At $d = 8, 9$ the ranks before this subtraction are 33, 44, respectively, which gives 32, 40 and verifies both remaining cases.

Here is a complete finite construction of all the columns used by OCF16. Start with $Z_0 = (1)$ and the invariant polynomial certificate $F_{0,1} = 1$. Suppose all earlier matrices Z_b have independent columns spanning $\bar{J}_b$, and their columns are $N_b \psi_b(F_{b,h})$ for invariant polynomials $F_{b,h}$. Form the candidate matrix

$$C_d = [m_{\nu,d} Z_{d-e}]_{\nu \in G_e, 2 \leq e \leq 5, e \leq d}. \quad (\text{OCF20})$$

Each candidate column has the explicit invariant certificate $x^\nu F_{d-e,h}$. OCF16 proves that C_d has column image $\bar{J}_d$ and rank j_d . Select j_d columns and j_d rows giving a nonzero determinant. Let Z_d be those columns, let I_d be those row positions, and let T_d be the complementary row positions in E_d^{strip} . Retain the corresponding invariant certificates. This completes the induction. When $j_d = 0$, use the empty matrix with zero columns, empty I_d , and $T_d = E_d^{\text{strip}}$.

Every selected determinant and every matrix entry is an explicit expression obtained from the original entries of $\mathbf{H}$ and the full a_{rs} . To cover all cases, order the finite choices of column and row subsets and select the first nonzero determinant; the corresponding chart asserts its nonvanishing and the vanishing of all preceding candidates. A nonzero determinant exists because the rank was proved in OCF19. At rank zero no determinant is needed. Together with the exhaustive leading-coefficient charts after OCF8, this gives a finite atlas for each specified k . It requires no generic coefficient premise and supplies no numerical zero test for an unevaluated analytic coefficient: each chart is specified by its exact minors.

Define $\rho_d : W_d \rightarrow \mathbb{C}^{T_d}$ and the full original coefficient map π_d by

$$\rho_d = \text{row}_{T_d} - (Z_d)_{T_d} (Z_d)_{I_d}^{-1} \text{row}_{I_d}, \quad \pi_d = \rho_d N_d. \quad (\text{OCF21})$$

At rank zero the second term is the zero map. If E_{T_d} inserts a column into its T_d positions in W_d , then

$$\begin{aligned} w &= Z_d (Z_d)_{I_d}^{-1} w_{I_d} + E_{T_d} \rho_d w, \\ \rho_d E_{T_d} &= I, \quad \ker \rho_d = \text{im } Z_d = \bar{J}_d, \\ \ker \pi_d &= J_d, \quad \pi_d S_d E_{T_d} = I_{\mathbb{C}^{T_d}}. \end{aligned} \quad (\text{OCF22})$$

For the first line, the subtraction on its right leaves zero I_d rows and exactly $\rho_d w$ on the complementary rows. This proves the first three claims. If $\pi_d f = 0$, then $N_d f \in \bar{J}_d$; choose $j \in J_d$ with $N_d j = N_d f$. Now $f - j \in \ker N_d = A\mathcal{B}_{d-8} \subseteq J_d$ by OCF17, so $f \in J_d$. The reverse inclusion follows immediately from $N_d J_d = \bar{J}_d$. The last equality uses $N_d S_d = I$. This proves every map and kernel in OCF22.

At the original degrees, these are the exact dimensions

k	n_k	j_k	$ T_k = n_k - j_k$
5	36	12	24
$k \geq 9, k = 4l + 1$	$16k - 48$	$8k - 32$	$8k - 16$.

(OCF23)

The original map $\Theta_k : R_k \xrightarrow{\sim} E_k^\vee$ in OQC7 sends the coefficient basis OCF2 to the original dual primary basis in the same order. Indeed its evaluation on $e^{wY_k}[1]$ is $\exp((2b-k)\gamma - i(2a-k)\delta)w$. The

corresponding eigenvalue of A_k is $\beta_{ab} = k/2 + (2a-k)\delta + i(2b-k)\gamma$; the eigenvalue of $Y_k = (A_k - kI/2)/i$ is $\omega_{ab} = (\beta_{ab} - k/2)/i = (2b-k)\gamma - i(2a-k)\delta$. These eigenvalues are distinct because $\delta > 0$ and $\gamma > 0$. Their exponentials are independent by the Vandermonde determinant, so the evaluation identifies the covectors exactly. By the original observation identity OQC8, $\Theta_k J_k = \text{im } \Lambda_k^\vee$. Therefore OCF22 supplies the specified isomorphism

$$E_k^\vee / \text{im } \Lambda_k^\vee \xrightarrow{\pi_k} \mathbb{C}^{T_k} \cong (\ker \Lambda_k)^\vee. \quad (\text{OCF24})$$

The displayed π_k uses exactly the original primary coefficient coordinates under Θ_k . Every dualization in this statement is complex-linear. No conjugation is inserted in this coefficient map; the original metric maps can subsequently be composed with it.

Finally, this construction is finite with explicitly bounded compressed matrices. OCF20 has n_d rows and $\sum_{e=2}^5 |G_e| j_{d-e}$ columns, with negative subscripts zero. Both dimensions grow at most linearly in d : for example $n_d \leq 16d + 1$ and the column number is at most $25(16d + 1)$. For $d \geq 13$ the exact column number is $200d - 1632$, obtained from $\sum_e |G_e| = 25$, $\sum_e e|G_e| = 104$, and $j_{d-e} = 8(d-e) - 32$. Only the preceding five graded bases are needed to form the next matrix. Each column is obtained from the at-most-36-coefficient formula OCF12, one exact convolution, and OCF6. A full original biform coefficient column, if used as temporary storage, has $(d+1)^2$ entries; every invariant relation matrix in OCF20 has the stated linear row and column counts. The invariant certificates can be retained as their products and parent-column references, so their expanded high-degree monomials need never be enumerated. The proof of their completeness is OCF11–16, and the exact original quotient and its right inverse are OCF21–24.

17.6 The same recursion with fourteen proved generators

The complete fixed set OCF9 can be reduced to the following fourteen specific members while preserving every map and image just proved:

e	$\mathcal{G}_e$	
2	$x_1 x_4, x_2 x_3$	
3	$x_1^2 x_3, x_1 x_2^2, x_2 x_4^2, x_3^2 x_4$	(OCF25)
4	$x_1^3 x_2, x_1 x_3^3, x_2^3 x_4, x_3 x_4^3$	
5	$x_1^5, x_2^5, x_3^5, x_4^5$	

Here is a complete proof. Call a positive-degree invariant monomial indecomposable when it is not the product of two positive-degree invariant monomials. The divisor proof preceding OCF11 shows that each indecomposable has degree at most five. If it contains both x_1, x_4 , it is their product, since otherwise division by this invariant quadratic leaves a nonconstant invariant monomial. The same assertion holds for x_2, x_3 . After these cases are removed, at most two variables occur: every three indices contain one of the two opposite pairs. One variable gives precisely its fifth power. The remaining two-element supports are $\{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}$. For positive exponents (a, b) with $a + b \leq 5$, the equation $a + 2b \equiv 0 \pmod{5}$ has exactly $(a, b) = (3, 1), (1, 2)$, and $a + 3b \equiv 0 \pmod{5}$ has exactly $(a, b) = (2, 1), (1, 3)$. For $\{2, 4\}$ divide the weights by 2 in the field $\mathbb{Z}/5\mathbb{Z}$, giving the first list; for $\{3, 4\}$ divide by 3, giving the second. These are exactly the eight mixed monomials of degrees three and four in OCF25. Every listed monomial has no proper invariant divisor, as is also seen from these lists and the excluded opposite pairs. This proves the classification. Factoring any invariant monomial repeatedly into indecomposables terminates because positive total degree strictly decreases. Hence OCF25 generates the entire invariant algebra.

Use $\mathcal{G} = \bigcup_{e=2}^5 \mathcal{G}_e$ in place of $\{x^\nu : \nu \in G\}$ in OCF12–16 and OCF20. The same factorization argument proves the same exact invariant image; all coefficient formulas, invariant certificates, pivot

charts, quotient maps and ranks remain valid. The new recursion has 2, 4, 4, 4 generators in degrees 2, 3, 4, 5, respectively. Thus for $d \geq 13$ its precise matrix dimensions and rank are

$$C_d^{(14)} \in \text{Mat}_{(16d-48) \times (112d-864)}(\mathbb{C}), \quad \text{rank } C_d^{(14)} = 8d - 32. \quad (\text{OCF26})$$

Indeed the number of generators is 14 and the sum of their degrees is $2 \cdot 2 + 4 \cdot 3 + 4 \cdot 4 + 4 \cdot 5 = 52$; substituting the already proved $j_{d-e} = 8(d-e) - 32$ gives $14(8d-32) - 8 \cdot 52 = 112d - 864$. At smaller degrees the exact number of columns is the same sum of the appropriate j_{d-e} from OCF19, with negative subscripts zero. All coefficients in this reduced recursion are still the original coefficients of OCF12–14.

Complete preceding Gamma reduction in the unchanged kernel

18 Independent derivation of the complete Gamma reduction for the original kernel

Date: 2026-09-20. Checked source: `JOINT_MINIMUM_RECEIVERS.tex`, JM21–JM28, in this directory. All constants and all displayed inequalities in those equations are accepted. The only textual correction is the reference to P5: P5 is the tail estimate, whereas the multiplier estimate used here is the Gamma multiplication estimate in its proof, written explicitly as C4 in `COERCIVITY_INDEPENDENT.md`.

The result retains the actual fixed kernel of the original observation. Its dimension $m_K=8k-16$ and the simple-packet formula $q=(k+1)^2$ are used on their original five-orbit domain. No kernel frame is selected or altered in this derivation. The asymptotic limit is along that domain with k congruent to 1 modulo 4.

18.1 1. Exact convolution density and its mass

Keep

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}, \quad M_\sigma = \sqrt{2\pi}.$$

Neither measure is divided by its mass. To check every convolution constant, use the Fourier convention

$$\hat{f}(t) = \int_{\mathbb{R}} e^{-ity} f(y) dy.$$

For any $a>0$ the beta integral gives

$$\frac{\Gamma(a + ix)\Gamma(a - ix)}{\Gamma(2a)} = \int_{-\infty}^{\infty} e^{ixv} (2 \cosh(v/2))^{-2a} dv. \quad (\text{KG1})$$

For completeness, multiply the two Euler integrals for Gamma and set $r=u+v$, $\theta=u/(u+v)$ in the positive quadrant. The Jacobian is r ; the r integral is $\Gamma(2a)$, and the remaining integral is $\int_0^1 \theta^{a+ix-1} (1-\theta)^{a-ix-1} d\theta$. Then set $z=\log(\theta/(1-\theta))$, whose derivative is $d\theta/dz=\theta(1-\theta)$. The resulting integrand is precisely the right side of KG1. Absolute convergence is ensured by $a>0$.

The function $(2 \cosh(v/2))^{-2a}$ and its first two derivatives are integrable. Its Fourier transform is therefore integrable: on a bounded interval it is bounded by the L1 norm, and outside that interval two integrations by parts give a constant times $|x|^{-2}$. Fourier inversion thus applies to KG1 as an ordinary absolutely convergent integral. After the exact substitution $y=2x$ it gives

$$\int_{\mathbb{R}} e^{-ity} \frac{|\Gamma(a + iy/2)|^2}{2\pi} dy = \frac{2\Gamma(2a)}{(2 \cosh t)^{2a}}. \quad (\text{KG2})$$

In particular

$$\hat{\sigma}(t) = M_\sigma (\cosh t)^{-1/2}, \quad \int \sigma = M_\sigma.$$

For the k -fold convolution, multiplication of Fourier transforms and KG2 with $a=k/4$ give

$$\sigma^{*k}(y) = \frac{M_\sigma^k 2^{k/2-1}}{\Gamma(k/2)} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \quad (\text{KG3})$$

Both sides are integrable, and their Fourier transforms agree; uniqueness of the Fourier transform proves the equality. Integrating either side shows that the mass is exactly $M_{\text{sigma}^k} = (2\pi)^{k/2}$.

Now put $\ell = (k-1)/4$. The Gamma recurrence gives

$$|\Gamma(1/4 + \ell + iy/2)|^2 = 4^{-\ell} \prod_{j=0}^{\ell-1} [y^2 + (2j + 1/2)^2] |\Gamma(1/4 + iy/2)|^2.$$

Since $2^{k/2-1} 4^{-\ell} = 2^{-1/2}$, KG3 becomes

$$\sigma^{*k}(y) = c_k \prod_{j < \ell} (y^2 + b_j^2) \sigma(y), \quad b_j = 2j + \frac{1}{2}, \quad c_k = \frac{(2\pi)^{k/2}}{\sqrt{2} \Gamma(k/2)}. \quad (\text{KG4})$$

This proves the exact JM21 constant. When $k=1$, $\ell=0$ and $c_1=1$; empty products therefore reproduce the original measure.

18.2 2. The whole polynomial comparison

Set $n=2q$, $a=a_{\{h,n\}}$, and $b=b_{\{h,n\}}$, with exactly the constants proved in P1–P2 and C16–C19 of COERCIVITY_INDEPENDENT.md. The finite positive form inequality on one-factor degree at most n is

$$a \|Q\|_{\sigma}^2 \leq \|Q\|_{w_h}^2 \leq b \|Q\|_{\sigma}^2.$$

Tensor products of positive matrices preserve this order, with factors a^k and b^k . For a scalar polynomial P of degree at most n , retain the physical total variable

$$S = s_1 + \cdots + s_k = k/2 + i(y_1 + \cdots + y_k) = c + iy, \quad c = k/2.$$

The polynomial $P(s_1 + \cdots + s_k)$ has separate degree at most n . Its tensor norm is exactly its scalar convolution norm. Thus

$$a^k \|P(c + i \cdot)\|_{\sigma^{*k}}^2 \leq \|P(c + i \cdot)\|_{w_h^{*k}}^2 \leq b^k \|P(c + i \cdot)\|_{\sigma^{*k}}^2. \quad (\text{KG5})$$

This retains the complete form, including cross terms between every pair of coefficients and every relation candidate.

The monic Gamma basis has norms and recurrence

$$\gamma_j = M_{\sigma} j! (1/2)_j, \quad p_0 = 1, \quad p_1 = y, \quad p_{j+1} = y p_j - j(j - 1/2) p_{j-1}. \quad (\text{KG6})$$

These formulas were checked from DLMF 18.19.8–9 and 18.23.7 in C1–C3 of COERCIVITY_INDEPENDENT.md, with the original measure retained. The two off-diagonal coefficients in its orthonormal basis are $\sqrt{j(j-1/2)}$. Consequently a polynomial Q of degree at most m satisfies

$$\|yQ\|_{\sigma} \leq 2(m+1) \|Q\|_{\sigma}. \quad (\text{KG7})$$

Indeed each of the two shifted coefficient vectors has norm at most $\sqrt{(m+1)(m+1/2)} \|Q\|$ or $\sqrt{m(m+1/2)} \|Q\|$; their sum is at most $2(m+1) \|Q\|$.

For $u_j = 2(n+j+1) + b_j$, KG7 and the triangle inequality give

$$\|(y + ib_j)Q\|_{\sigma} \leq u_j \|Q\|_{\sigma} \quad \text{when } \deg Q \leq n + j.$$

After j preceding factors, the degree is at most $n+j$. Iteration therefore gives exactly

$$\left\| \prod_{j<\ell} (y + ib_j) P(c + iy) \right\|_{\sigma}^2 \leq \left(\prod_{j<\ell} u_j^2 \right) \|P(c + i \cdot)\|_{\sigma}^2. \quad (\text{KG8})$$

The matching lower bound uses the pointwise inequality $\prod (y^2 + b_j^2) \geq \prod b_j^2$. Combining KG4, KG5, and KG8 proves

$$\boxed{\alpha_k \|P(c + i \cdot)\|_{\sigma}^2 \leq \|P(c + i \cdot)\|_{w_h^{*k}}^2 \leq \beta_k \|P(c + i \cdot)\|_{\sigma}^2,} \quad (\text{KG9})$$

$$\alpha_k = a^k c_k \prod_{j<\ell} b_j^2, \quad \beta_k = b^k c_k \prod_{j<\ell} u_j^2.$$

This is JM22. The source measure on the right and left of KG9 is the order-one Gamma measure in the variable y , while the physical scalar coordinate remains $S=c+iy$ throughout.

The finite logarithmic width is therefore

$$L_k^{\text{form}} = \log(\beta_k/\alpha_k) = k \log(b/a) + 2 \sum_{j<\ell} \log(u_j/b_j) \geq 0. \quad (\text{KG10})$$

The scalar c_k cancels in this ratio; it was included in both bounds before the cancellation. There is also an explicit upper bound for the second term. Since

$$\frac{u_j}{b_j} = 2 + \frac{2n + 3/2}{2j + 1/2} \leq 4n + 5,$$

one has

$$0 \leq L_k^{\text{form}} \leq k \log(b/a) + 2\ell \log(4n + 5) = O_h(k \log^2(q + 2)). \quad (\text{KG11})$$

This verifies JM23, with no hidden scalar mass and no root-gap estimate.

18.3 3. Quotient first, actual kernel restriction second

Let $E = C[S]/(\chi_k(S))$ be the unchanged original scalar quotient of dimension q . For every $q-1 \leq N \leq 2q$, let $J_N: P_{\leq N} \rightarrow E$ be the original remainder map. For each v in E its affine fibre is nonempty. The complete physical minimum and Gamma minimum are

$$v^* G_N v = \min_{J_N P = v} \|P(c + i \cdot)\|_{w_h^{*k}}^2, \quad v^* G_N^{\sigma} v = \min_{J_N P = v} \|P(c + i \cdot)\|_{\sigma}^2. \quad (\text{KG12})$$

These minima exist because the spaces are finite-dimensional and their norm matrices are positive definite. Apply KG9 to every polynomial in the identical fibre. Taking the minima of the three forms gives

$$\alpha_k G_N^{\sigma} \preceq G_N \preceq \beta_k G_N^{\sigma}. \quad (\text{KG13})$$

No relation vector has been omitted, and no polynomial representative has been fixed before minimizing.

Let the original fixed kernel be $K = \ker \Lambda_0$, and let $I_K: C^{\infty} \rightarrow E$ be its actual injective frame map. Set

$$H_{K,N} = I_K^* G_N I_K, \quad H_{K,N}^{\sigma} = I_K^* G_N^{\sigma} I_K.$$

Restriction of KG13 is a congruence by that same I_K . Both restricted matrices are positive definite. The eigenvalues of $(H_{\{K,N\}}^{\sigma})^{-1/2} H_{\{K,N\}} (H_{\{K,N\}}^{\sigma})^{-1/2}$ lie in $[\alpha_k, \beta_k]$, so the product of its m_K eigenvalues yields

$$m_K \log \alpha_k \leq e_N \leq m_K \log \beta_k, \quad e_N = \log \det H_{K,N} - \log \det H_{K,N}^{\sigma}. \quad (\text{KG14})$$

This is JM24. It is the restriction of the full quotient metric; it is not obtained by retaining selected coefficient directions before the quotient minimum.

Use the four original signs

$$\mathcal{R}e = e_{q-1} + e_q - e_{2q-1} - e_{2q}.$$

All four e_N belong to the same interval of length $m_K L_k^{\text{form}}$. Pair the first term with the third and the second with the fourth. Each difference has absolute value at most that length; hence

$$\left| \mathcal{R} \log \det H_{K,N} - \mathcal{R} \log \det H_{K,N}^{\sigma} \right| \leq 2m_K L_k^{\text{form}}. \quad (\text{KG15})$$

No monotonicity of the errors e_N is assumed. This verifies the factor 2 in JM25. The common scalar $m_K \log c_k$ in each e_N cancels in this four-sign expression as well.

On the actual simple-packet domain,

$$2m_K L_k^{\text{form}} = O_h(k^2 \log^2(q+2)), \quad \frac{k^2 \log^2(q+2)}{kq} = \frac{k \log^2((k+1)^2 + 2)}{(k+1)^2} \rightarrow 0. \quad (\text{KG16})$$

Thus the error is $o_h(kq)$, as stated, while retaining all roots and the entire original kernel.

18.4 4. Exact covariance in the transported quotient

To specify the coordinate change completely, if χ_k is monic of degree q , set

$$Q(y) = i^{-q} \chi_k(c + iy).$$

This is monic and retains every original root with its full multiplicity; the root λ is transported to $(\lambda - c)/i$. The substitution

$$T : E \rightarrow E_y = \mathbb{C}[y]/(Q), \quad T[P] = [P(c + iy)] \quad (\text{KG17})$$

is an isomorphism, with inverse substitution $y = (s - c)/i$. Its coefficient matrix is triangular with diagonal $1, i, \dots, i^{q-1}$. The transported kernel frame is $I_{\{K,y\}} = T I_K$. The matrices transform as

$$G_{N,y}^{\sigma} = (T^{-1})^* G_N^{\sigma} T^{-1}, \quad C_{N,y}^{\sigma} = T C_N^{\sigma} T^*.$$

Consequently $I_{\{K,y\}}^* G_{\{N,y\}}^{\sigma} I_{\{K,y\}} = I_K^* G_N^{\sigma} I_K$ exactly. No coordinate determinant or kernel-frame factor is lost.

In E_y use the standard coefficient frame of degree below q and define

$$v_j = \text{rem}_Q p_j,$$

where p_j and γ_j are exactly KG6. The orthonormal source basis is $p_j / \sqrt{\gamma_j}$, for all $0 \leq j \leq N$; its remainder matrix has columns $v_j / \sqrt{\gamma_j}$. Thus its complete covariance is

$$C_{N,y}^{\sigma} = \sum_{j=0}^N \frac{v_j v_j^*}{\gamma_j}. \quad (\text{KG18})$$

For $j < q$, p_j already has degree below q , so $v_j = p_j$. The first q columns are triangular with diagonal one. They have full rank, proving $C_{\{N,y\}}^{\sigma} > 0$ for every $N \geq q-1$.

For a direct proof of the minimum formula, let B be this remainder matrix on orthonormal source coefficients. Since $BB^* = C > 0$, the coefficient vector $B^* C^{-1} v$ has value v and norm squared $v^* C^{-1} v$. Every other coefficient vector with value v differs from it by $\ker B$, which is orthogonal to $\text{im } B$. Its squared norm therefore equals that minimum plus the squared norm of the difference. It follows that

$$G_{N,y}^{\sigma} = (C_{N,y}^{\sigma})^{-1}, \quad \boxed{H_{K,N}^{\sigma} = I_{K,y}^* (C_{N,y}^{\sigma})^{-1} I_{K,y}}. \quad (\text{KG19})$$

These are JM26–JM27 with the coordinate transport specified. The inverse is taken on the complete q -dimensional value space before restriction to the actual m_K -dimensional kernel. In general $I_K^* C^{-1} I_K$ is different from $(I_K^* C I_K)^{-1}$; only the former is proved and used here. Every column through degree N is present in KG18, so every earlier scalar relation enters the minimum.

18.5 5. The actual change to the joint kernel metric

Keep the same section at all four cutoffs, as in J3–J11. The already proved joint form enclosure and nesting give

$$0 \leq \mathcal{R} \log \det H_{K,N}^J \leq 2m_K W, \quad W = k \log(\tau_{\eta} \Lambda_{h,2q}). \quad (\text{KG20})$$

By definition $\Delta_{\{K,N\}} = \log \det H_{\{K,N\}} - \log \det H_{\{K,N\}}^J$. Combine KG15 with both endpoints of KG20 to obtain

$$\boxed{\begin{aligned} \mathcal{R} \log \det H_{K,N}^{\sigma} - 2m_K (L_k^{\text{form}} + W) &\leq \mathcal{R} \Delta_{K,N} \\ &\leq \mathcal{R} \log \det H_{K,N}^{\sigma} + 2m_K L_k^{\text{form}}. \end{aligned}} \quad (\text{KG21})$$

This is JM28, with its signs and asymmetric error terms unchanged. At the declared polynomial tolerances, $W = O_h(k \log^2(q+2))$; thus both errors are $O_h(kq)$ by KG16. More specifically, for the actual simple one-factor divisor, $d_{\max} = 0$, $\tau_{\eta} = L_h$, and W is independent of η altogether. The polynomial-tolerance statement is therefore valid, and on this simple domain the same error bound holds for every positive η at most one when the section is fixed across cutoffs.

The derived expression KG19 is the full finite Gamma receiver needed for the next kernel calculation. The reduction preserves a possible kq coefficient of its four-return, but does not assign that coefficient without evaluating the determinant asymptotics in the actual frame.

18.6 Reading and use record

The primary task source actually read was the displayed section of `JOINT_MINIMUM_RECEIVERS.tex` containing JM21–JM28. The prior independent proof `COERCIVITY_INDEPENDENT.md` supplies P2 and the exact Gamma basis and multiplier estimates; its original DLMF TeX files and precise reading record remain in `coercivity_sources/` and its Section 11. The convolution identity and every new error estimate were derived above from the exact Gamma Euler integrals and those already checked inputs. No additional literature, numerical experiment, rendering, build, or publication is represented as performed.

Complete preceding arithmetic coercivity and tensor minimum proof

19 Independent proof of the polynomial and tensor coercivity estimates

Date: 2026-09-20. Scope: equations P1–P11 and J1–J11 in the supplied COMPLETE_PROOFS(2).md, Sections 1–2. This file gives the complete independent derivation within that scope. Sections 3–5 have not been audited here.

19.1 Acceptance and precise scope

All numerical constants in P1–P11 are valid. J1–J11 are valid for the original nested sources, a nonempty fixed divisor, positive fixed jet weights, and a single section fixed at the four cutoffs. The formulas using the inverse of the scalar minimum metric require its attained domain, namely $N \geq q-1$. The joint minimum itself exists for every $N \geq 0$ because the section is onto.

One sentence requires a scope correction. The last paragraph of Section 2 extends all results to intermediate sources inside the same polynomial box. The common metric bounds extend to every such source containing the section. The nonnegative lower bound in J11 additionally requires that these sources increase with N . Without this nesting, the enclosure gives the valid bound $|\mathbf{R} \log \det| \leq 2 \mathbf{r}_X \mathbf{W}_k$, but it does not prove a nonnegative return. The actual sources $X_N + Y_{\{k, \eta\}}$ are nested, so their stated J11 is valid.

For general multiplicities the exact tolerance cost is retained below. Nothing here removes $\log(1/\eta)$ from that cost when η is allowed to shrink arbitrarily. For simple factors, the metric has no higher jets and the cost does not depend on η .

The displayed maxima over roots and the degree $n_0=2d-1$ in the supplied formulas presuppose a nonempty divisor. This is the domain used below. No convention for an empty root set is silently inserted.

19.2 1. Original spaces, coordinates, and physical unit

Let Z be a finite nonempty set of distinct nontrivial zeros of ζ , and let each m_ρ be its complete order in $2\mathbf{x}$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \sum_{\rho} m_\rho, \quad d_\rho = m_\rho - 1, \quad d_{\max} = \max_{\rho} d_\rho.$$

Fix $a_\rho > 0$. The physical line is $s=1/2+iy$. In all norm integrals,

$$\|P\|_{w_h}^2 = \int_{\mathbb{R}} |P(1/2 + iy)|^2 \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi} dy,$$

$$\|P\|_{\sigma}^2 = \int_{\mathbb{R}} |P(1/2 + iy)|^2 \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy.$$

When a polynomial is displayed as a function of y , it means the exact substitution $Q(y)=P(1/2+iy)$. No density is divided by its mass.

The coefficient quotient is $A_h = \mathbb{C}[s]/(h)$, represented by polynomials of degree below d . The divided-jet map is

$$H_h : A_h \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad [P] \longmapsto \left(\sum_{j=0}^{d_\rho} \frac{P^{(j)}(\rho)}{j!} t_\rho^j \right)_\rho.$$

This is an isomorphism: its kernel consists precisely of multiples of h , and its source and target both have dimension d . In these jets define

$$D_{h,\eta}[P] = \sum_{\rho,j} a_\rho(j!)^2 \binom{d_\rho}{j} \eta^{2j} \left| \frac{P^{(j)}(\rho)}{j!} \right|^2, \quad 0 < \eta \leq 1, \quad T_\eta = H_h^* D_{h,\eta} H_h.$$

Here and below a positive matrix also denotes its quadratic form when the argument is written in brackets.

Put $f_h = 2xi/h$. It is entire and $f_h(\rho)$ is nonzero at every selected root, because the complete order was removed. Multiplication by its truncated Taylor series defines the invertible jet map U_h . Every coefficient is retained:

$$(U_h u)_{\rho,j} = \sum_{r=0}^j \frac{f_h^{(r)}(\rho)}{r!} u_{\rho,j-r}.$$

With the physical metric $D_{\text{phys}} = (U_h^{-1})^* D_{h,1} U_h^{-1}$,

$$\|U_h u\|_{D_{\text{phys}}}^2 = u^* D_{h,1} u.$$

The tensor identity is obtained by taking tensor powers of this exact identity. Thus the later value constraint βv and the physical constraint $U_h^{\otimes k} \beta v$ have equal specified norms; the physical unit and its derivatives have not been discarded.

19.3 2. Gamma polynomial basis, mass, and multiplication bound

The Meixner–Pollaczek formulas used are DLMF 18.23.7, 18.19.8, and 18.19.9. Substitute $\lambda = 1/4$, $\phi = \pi/2$, $x = y/2$, and define

$$p_m(y) = m! P_m^{(1/4)}(y/2; \pi/2).$$

Then

$$\sum_{m \geq 0} \frac{p_m(y)}{m!} z^m = (1 + z^2)^{-1/4} \exp(y \arctan z), \quad |z| < 1. \quad (C1)$$

The leading coefficient in y is one, as is also immediate by selecting the highest y power in C1. The exact change $dy = 2dx$ in DLMF's orthogonality gives

$$\langle p_m, p_l \rangle_\sigma = \delta_{ml} \gamma_m, \quad \gamma_m = \sqrt{2\pi} m! (1/2)_m. \quad (C2)$$

Indeed the DLMF norm of P_m is $2 \pi \Gamma(m+1/2) / (\sqrt{2} m!)$; multiplying by $(m!)^2 / \pi$ gives C2. In particular $M_{\sigma=0} = \sqrt{2\pi}$, exactly as in P10.

Differentiate C1 in z . Its right side satisfies

$$(1 + z^2) \partial_z C1 = (y - z/2) C1.$$

Comparison of coefficients gives, with $p_{-1}=0$,

$$yp_m = p_{m+1} + m(m-1/2)p_{m-1}. \quad (C3)$$

Thus in the orthonormal basis $e_m = p_m / \sqrt{\gamma_m}$, the off-diagonal coefficient is $\alpha_m = \sqrt{m(m-1/2)}$. If $\deg Q \leq n$, the two shifted coefficient vectors in yQ have norms at most $\alpha_{n+1} \|Q\|$ and $\alpha_n \|Q\|$. The triangle inequality yields

$$\|yQ\|_\sigma \leq (\alpha_{n+1} + \alpha_n) \|Q\|_\sigma \leq 2(n+1) \|Q\|_\sigma. \quad (C4)$$

For $R=8(n+1)$, integration of $1_{\{|y|>R\}} \leq y^2/R^2$ now gives

$$\int_{|y|>R} |Q(y)|^2 \sigma(y) dy \leq \frac{1}{16} \|Q\|_\sigma^2. \quad (C5)$$

This verifies P5, including its degree and constant.

19.4 3. Legendre localization and the excluded mass

DLMF 5.7.6 gives, for $a>0$ and $t \geq 0$,

$$\operatorname{Im} \psi(a+it) = \frac{t}{a^2+t^2} + \sum_{j \geq 1} \frac{t}{(j+a)^2+t^2}.$$

The first summand is at most $1/(2a)$. The remaining summand is decreasing as a function of $j \geq 0$, so its sum for $j \geq 1$ is at most

$$\int_0^\infty \frac{t}{(x+a)^2+t^2} dx \leq \pi/2.$$

At $a=1/4$ this proves $|\operatorname{Im} \psi(a+it)| \leq 2+\pi/2 < 4$. Differentiating the actual density, with its factor $y/2$, gives

$$\frac{d}{dy} \log \sigma(y) = -\operatorname{Im} \psi(1/4 + iy/2).$$

Complex conjugation handles negative y . Therefore, for $|u-y| \leq 1$,

$$\sigma(y) \leq e^4 \sigma(u). \quad (C6)$$

Let L_m be the Legendre polynomial on $[-1,1]$, with $\int_{-1}^1 L_m^2 = 2/(2m+1)$. Its value at zero is 0 for odd m ; for $m=2j$ it is $(-1)^j \binom{2j}{j}/4^j$, of modulus at most 1. Expand the complete complex polynomial $Q(y+t)$ in the orthonormal functions $\sqrt{(2m+1)/2} L_m(t)$. Cauchy-Schwarz at $t=0$ gives

$$|Q(y)|^2 \leq \left(\sum_{m=0}^n \frac{2m+1}{2} |L_m(0)|^2 \right) \int_{y-1}^{y+1} |Q(u)|^2 du \leq \frac{(n+1)^2}{2} \int_{y-1}^{y+1} |Q(u)|^2 du.$$

Multiply by $\sigma(y)$, use C6 inside the integral, and extend that integral to the whole line. This proves P4 with exactly

$$B_n = e^4 (n+1)^2 / 2. \quad (C7)$$

Consequently every measurable excluded set E of length at most $1/(32B_n)$ has $\int_E |Q|^2 \sigma \leq \|Q\|_\sigma^2 / 32$.

19.5 4. The zeta bound and all disk constants

Set $g(s)=(s-1)\zeta(s)$, which is entire. Taking $n=1$ in DLMF 25.2.10 and integrating its periodic B_{-3} remainder using $B_{-4}'=4B_{-3}$ gives

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{s}{12} - \frac{s(s+1)(s+2)}{720} - \frac{(s)_4}{24} \int_1^\infty \tilde{B}_4(x) x^{-s-4} dx. \quad (C8)$$

Initially this integration is valid on the common convergence domain; the resulting absolutely convergent integral extends the identity to $\operatorname{Re} s > -3$. The sign and coefficient of its boundary term follow from $B_{-4}(1)=-1/30$ and $[-B_{-4}(1)/4]=1/120$ in the integration of B_{-3} . Since $B_{-4}(t)=t^2(1-t)^2-1/30$ on $[0,1]$, its range lies in $[-1/30, 7/240]$; hence $\sup |B_{-4}|=1/30$.

If $\operatorname{Re} s \geq -2$, the absolute value of the integral in C8 is at most $1/30$. Put $t=|s|$, $T=t+4 \geq 4$. Multiplication by $s-1$ yields the explicit estimate

$$\begin{aligned} |g(s)| &\leq 1 + \frac{t+1}{2} + \frac{t(t+1)}{12} + \frac{t(t+1)^2(t+2)}{720} + \frac{t(t+1)^2(t+2)(t+3)}{720} \\ &\leq 1 + T/2 + T^2/12 + T^4/720 + T^5/720 < T^5. \end{aligned} \quad (C9)$$

For the last inequality, division by T^5 bounds the right side by $1/1024+1/512+1/768+1/2880+1/720 < 1$. At the removable point $s=1$ the same inequality follows by continuity. This verifies P6.

The Euler product in $\operatorname{Re} s > 1$ gives

$$|\zeta(2+ij)^{-1}| \leq \prod_p (1+p^{-2}) \leq \prod_p (1-p^{-2})^{-1} = \zeta(2) < 2.$$

The last strict inequality also follows directly from the sum and the integral comparison $\sum_{m \geq 2} m^{-2} < \int_1^\infty x^{-2} dx = 1$. Since $|1+ij| \geq 1$, $|g(2+ij)| > 1/2$.

Fix an integer j with $|j| \leq R$, and put $f_j(z)=g(2+ij+z)$. On $|z| \leq 4$, $\operatorname{Re}(2+ij+z) \geq -2$ and $|2+ij+z|+4 \leq R+10$. Therefore C9 gives

$$|f_j(z)| \leq (R+10)^5 \leq M_n := 4(R+14)^5. \quad (C10)$$

Let $L_n = \log(2M_n)$, and let N_j be the number of zeros a of f_j in $|a| < 3$, counted with multiplicity. Jensen's formula at radii tending to 4 from below gives

$$N_j \log(4/3) \leq \log M_n - \log |f_j(0)| < L_n.$$

No zero occurs at zero. Thus

$$N_j \leq K_n = \lceil L_n / \log(4/3) \rceil. \quad (C11)$$

This reasoning allows zeros on the circle of radius 4 by taking the limit through radii avoiding their moduli.

Form the finite product

$$B_j(z) = \prod_{|a| < 3} \frac{3(z-a)}{9-\bar{a}z}, \quad F_j(z) = f_j(z)/B_j(z),$$

with multiplicities. All apparent poles of F_j at the listed zeros are removable, and there are no zeros of F_j in $|z| < 3$. On $|z|=3$ every factor has modulus one, so the maximum principle gives $|F_j| \leq M_n$ on the disk. Also $|B_j(0)| \leq 1$, hence $|F_j(0)| > 1/2$.

The nonnegative harmonic function $H = \log(M_n/|F_j|)$ has $H(0) < L_n$. To apply Harnack without any boundary regularity assumption, use every radius r with $2 < r < 3$ and then let r increase to 3. At $|z| \leq 2$,

$$H(z) \leq \lim_{r \uparrow 3} \frac{r+2}{r-2} H(0) \leq 5L_n.$$

Thus

$$|F_j(z)| \geq M_n e^{-5L_n} = \frac{1}{32M_n^4}, \quad |z| \leq 2. \quad (\text{C12})$$

For each removed zero, $|\mathbf{9-bar(a)z}| \leq 15$ on $|z| \leq 2$. Hence

$$\left| \frac{3(z-a)}{9-\bar{a}z} \right| \geq |z-a|/5. \quad (\text{C13})$$

The critical-line cell $j-1/2 \leq y \leq j+1/2$ has relative coordinate $z = -3/2 + i(y-j)$, whose modulus is at most $\sqrt{5/2} < 2$. Let

$$m_n = 2R + 1, \quad \epsilon_n = (64B_n m_n K_n)^{-1}, \quad c_n = (\epsilon_n/5)^{K_n} / (32M_n^4).$$

The number $\epsilon_n/5$ is below one. Outside the ϵ_n disks about the listed zeros, C11–C13 imply $|g(1/2+iy)| \geq c_n$. Each disk cuts an interval of length at most $2\epsilon_n$ from the real y line. Across all m_n cells the total excluded length is at most

$$2m_n K_n \epsilon_n = 1/(32B_n). \quad (\text{C14})$$

Counting a disk or an overlapping interval more than once can only increase this upper bound. By C5 and C7, the retained part of $[-R, R]$ carries Gamma polynomial mass at least $1-1/16-1/32=29/32$ of the full mass, and therefore at least one half. This establishes P7 with precisely the stated constants.

19.6 5. Lower and upper arithmetic comparisons

The defining identity $2\chi(s) = s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$ gives on the physical line

$$\frac{w_h(y)}{\sigma(y)} = \frac{|s|^2}{\sqrt{\pi}} \left| \frac{g(s)}{h(s)} \right|^2, \quad s = 1/2 + iy. \quad (\text{C15})$$

This formula uses the entire removable value of g/h at selected roots. Away from $h=0$ it is exactly P8 in its displayed fraction form.

Let $R_h = 2 + \max_{\rho} |\rho - 1/2|$. On $|y| \leq R$,

$$|h(1/2 + iy)| \leq (R + R_h)^d, \quad |s|^2 \geq 1/4.$$

The retained set contains no zero of g . In particular h is nonzero there, so division is legitimate. C15 and the retained half-mass yield

$$\|P\|_{w_h}^2 \geq \frac{c_n^2}{8\sqrt{\pi}(R + R_h)^{2d}} \|P\|_{\sigma}^2 = a_{h,n} \|P\|_{\sigma}^2. \quad (\text{C16})$$

For the upper comparison, g/h is entire because h divides the actual nontrivial-zero divisor of g . Consider the rectangle $-1 \leq \operatorname{Re} s \leq 2$, $|\operatorname{Im} s| \leq 2R_h$. On its vertical sides, every $|\mathbf{s-rho}| \geq 1$, since $0 < \operatorname{Re} \rho < 1$. On its horizontal sides,

$$|s - \rho| \geq 2R_h - |\operatorname{Im} \rho| \geq 2R_h - (R_h - 2) = R_h + 2 > 1.$$

Thus $|h| \geq 1$ on the entire boundary. Also $|s| + 4 \leq 2R_h + 6$ there. C9 and the maximum principle applied to the entire g/h prove

$$|g/h| \leq (2R_h + 6)^5$$

throughout the rectangle, including all removable values. On its critical line segment, C15 consequently gives

$$w_h/\sigma \leq [(2R_h)^2 + 1](2R_h + 6)^{10}. \quad (\text{C17})$$

For $|y| \geq 2R_h + 4$, triangle inequalities give

$$|s - \rho| \geq |y| - |\rho - 1/2| \geq |y|/2, \quad |h(s)| \geq (|y|/2)^d \geq 1,$$

and $|s| \leq 2|y|$, $|s| + 4 \leq 3|y|$. Hence C9 and C15 give

$$w_h/\sigma \leq (4 \cdot 3^{10}/\sqrt{\pi})|y|^{12} \leq 2^{24}(1 + y^2)^6. \quad (\text{C18})$$

Combining C17 and C18 proves P9 with exactly

$$C_h = \max\{2^{24}, [(2R_h)^2 + 1](2R_h + 6)^{10}\}.$$

For every polynomial of degree at most m , C4 implies $\|(y+i)Q\|_{\sigma} \leq (2m+3)\|Q\|_{\sigma}$. Iteration at $m=n, n+1, \dots, n+5$ gives

$$\|(y+i)^6 Q\|_{\sigma}^2 \leq \prod_{r=0}^5 (2n+2r+3)^2 \|Q\|_{\sigma}^2 \leq (2n+15)^{12} \|Q\|_{\sigma}^2.$$

Because $|(y+i)^6| \leq (1+y^2)^6$, P9 proves the upper half of P2 with the unchanged $b_{\{h,n\}} = C_h(2n+15)^{12}$.

Now $K_n = O(\log(n+2))$, $\log M_n = O(\log(n+2))$, and

$$-\log \epsilon_n = \log 64 + \log B_n + \log m_n + \log K_n = O(\log(n+2)).$$

The explicit logarithm of $a_{\{h,n\}}$ is

$$-\log a_{h,n} = 2K_n \log(5/\epsilon_n) + 2 \log 32 + 8 \log M_n + \log(8\sqrt{\pi}) + 2d \log(R + R_h). \quad (\text{C19})$$

It is $O_h(\log^2(n+2))$. Also $\log b_{\{h,n\}} = \log C_h + 12 \log(2n+15) = O_h(\log(n+2))$. This proves P3 with no change to the constants.

19.7 6. The complete divided-jet bound

Let $r = n/(n+1)$ and $u_n = (1/2) \log(2n+1)$. On $|z| = r$ the power series for arctan gives

$$|\arctan z| \leq \sum_{l \geq 0} \frac{r^{2l+1}}{2l+1} = \operatorname{arctanh} r = u_n.$$

Further,

$$|1 + z^2|^{-1/4} \leq (1 - r^2)^{-1/4} \leq (n+1)^{1/4}, \quad r^{-m} \leq (1 + 1/n)^n \leq e \quad (m \leq n).$$

Write $Z_h = \max_{\rho} |\rho - 1/2|$ and $y_{\rho} = (\rho - 1/2)/i$, so $|y_{\rho}| \leq Z_h$. Cauchy's coefficient estimate applied to the j -th divided y derivative of C1 gives

$$\left| \frac{p_m^{(j)}(y_{\rho})}{j!} \right| \leq m! e(n+1)^{1/4} (2n+1)^{Z_h/2} \frac{u_n^j}{j!}. \quad (\text{C20})$$

The s derivative of $p_m((s-1/2)/i)$ contributes the exact phase i^{-j} , of modulus one. The inequality $m!/(1/2)_m \leq 2m+1$ holds by induction: its ratio at the next step is $(m+1)/(m+1/2)$, and $(2m+1)(m+1)/(m+1/2) = 2m+2 \leq 2m+3$.

Expand Q in the orthonormal basis $e_0, \dots, e_n$ and apply Cauchy–Schwarz to its j -th divided derivative at y_ρ . C2 and C20 give

$$\left| \frac{P^{(j)}(\rho)}{j!} \right|^2 \leq \|P\|_\sigma^2 \frac{e^2}{M_\sigma} (n+1)^{1/2} (2n+1)^{Z_h} \frac{u_n^{2j}}{(j!)^2} \sum_{m=0}^n (2m+1).$$

The sum is $(n+1)^2 \leq (n+1)(2n+1)$. Thus

$$\left| \frac{P^{(j)}(\rho)}{j!} \right|^2 \leq \|P\|_\sigma^2 \frac{e^2}{M_\sigma} (n+1)^{3/2} (2n+1)^{Z_h+1} \frac{u_n^{2j}}{(j!)^2}. \quad (C21)$$

Multiply by the original weights $a_\rho (j!)^2 \text{binom}(d_\rho, j)$ and sum over all roots and all jet orders. The factorials cancel exactly; the remaining sum over j is $(1+u_n^2)^{d_\rho}$. Therefore

$$\|J_h P\|_{D_{h,1}}^2 \leq J_{h,n} \|P\|_\sigma^2, \quad J_{h,n} = \frac{e^2}{M_\sigma} (n+1)^{3/2} (2n+1)^{Z_h+1} \sum_\rho a_\rho (1+u_n^2)^{d_\rho}. \quad (C22)$$

Combining C22 and C16 gives P11 with $\text{Lambda}_{\{h,n\}} = \max\{1, J_{\{h,n\}}/a_{\{h,n\}}\}$. The fixed finite root set and fixed weights give $\log^+ J_{\{h,n\}} = O_h(\log(n+2))$; C19 therefore proves $\log \text{Lambda}_{\{h,n\}} = O_h(\log^2(n+2))$.

19.8 7. Tensor forms and exact collision coefficients

Let V_n be the full polynomial space of degree at most n with its w_h inner product, and let $J: V_n \rightarrow A_h$ be the remainder map. C22 and C16 say that the finite operator

$$J: (V_n, \|\cdot\|_{w_h}) \longrightarrow (A_h, T_1)$$

has squared operator norm at most $\text{Lambda}_{\{h,n\}}$. Tensoring its singular value decomposition shows that the squared norm of $J^{\otimes k}$ is at most $\text{Lambda}_{\{h,n\}}^k$. This acts on the full Hilbert tensor product $V_n^{\otimes k}$, which is precisely the space of polynomials of separate degree at most n . Its norm is the original product-measure norm. This proves J1 for every coefficient vector, including all cross terms.

For clarity the cyclic source and its complete collision coefficients can be obtained directly. For an ordered tuple $\text{boldrho} = (\rho_1, \dots, \rho_k)$, put

$$\kappa(\rho) = \sum_i \rho_i, \quad D(\rho) = \sum_i d_{\rho_i}, \quad A(\rho) = \prod_i a_{\rho_i}.$$

Let K be the set of distinct attained κ , and set

$$e_\kappa = 1 + \max_{\kappa(\rho)=\kappa} D(\rho), \quad \chi(S) = \prod_{\kappa \in K} (S - \kappa)^{e_\kappa}, \quad E = \mathbb{C}[S]/(\chi), \quad q = \deg \chi.$$

The map

$$\beta: E \longrightarrow A_h^{\otimes k}, \quad [P(S)] \longmapsto [P(s_1 + \dots + s_k)] \quad (C23)$$

is well-defined and injective. In the primary factor for a tuple, $s_i = \rho_i + t_i$ and $t_i^{d_{\rho_i}+1} = 0$. The element $t_1 + \dots + t_k$ has nilpotence order $D+1$: its D -th power is the nonzero multiple $\frac{D!}{\prod_i d_{\rho_i}!} \prod_i t_i^{d_{\rho_i}}$, and

every monomial of degree $D+1$ vanishes. For each center the largest such D therefore imposes exactly the exponent e_{κ} . Taking all tuple factors shows that the kernel of the substitution from $C[S]$ is exactly (χ) .

Write $v_{\{\kappa, j\}} = P^{\{(j)\}}(\kappa)/j!$. In the tuple component of βv , the coefficient of $\prod_i t_i^{j_i}$, with $\sum j_i = j$, is

$$\frac{j!}{\prod_i j_i!} v_{\kappa, j}.$$

Its squared coefficient times its exact product metric weight is

$$(j!)^2 |v_{\kappa, j}|^2 A(\rho) \prod_i \binom{d_{\rho_i}}{j_i}.$$

The sum over the j_i is the coefficient of x^j in $\prod_i (1+x)^{d_{\rho_i}} = (1+x)^D$. Thus, with zero binomial coefficients when $j > D$,

$$S_{\kappa, j} = \sum_{\kappa(\rho) = \kappa} A(\rho) \binom{D(\rho)}{j}, \quad G_{k,1}(v) = \sum_{\kappa, j} (j!)^2 S_{\kappa, j} |v_{\kappa, j}|^2. \quad (C24)$$

Each $S_{\{\kappa, j\}}$ is positive on the stated j range, since a tuple of maximum D contributes positively. Every ordered tuple, including tuples with unequal lengths at a common center, is included. With $D_{\{h, \eta\}}$ in every factor, the product contributes $\eta^{2\sum j_i} = \eta^{2j}$; hence

$$G_{k, \eta}(v) = \sum_{\kappa, j} \eta^{2j} (j!)^2 S_{\kappa, j} |v_{\kappa, j}|^2 \preceq G_{k,1}(v). \quad (C25)$$

For every full polynomial F with $J^{\{\text{tensor } k\}} F = \beta v$, J_1 and $C24$ give the exact J_2 lower bound $\|F\|^2 \geq \Lambda_{\{h, n\}}^{\{-k\}} G_{\{k, 1\}}(v)$.

19.9 8. The section and the complete minimum

Set $n_0 = 2d-1$, and use the original coefficient frame on $V_{\{n_0\}}$. Its source Gram H is positive definite: w_h is positive except at a discrete set, so the squared norm of a nonzero polynomial is positive. The remainder matrix J has rank d . Its kernel consists of hQ with $\deg Q \leq d-1$, so it has dimension d . Define

$$G_0 = (JH^{-1}J^*)^{-1}, \quad R_0 = H^{-1}J^*G_0.$$

Then $JR_0 = I$ and $R_0^*HR_0 = G_0$. Choose W with d columns spanning $\ker J$ and $W^*HW = I_d$. Direct multiplication gives $R_0^*HW = G_0JW = 0$.

Let $L_h = 1 + \|T_1^{-1/2}G_0T_1^{-1/2}\|$ and $\tau_\eta = L_h \eta^{2d_{\max}}$. The exact jet weights imply

$$\eta^{2d_{\max}} T_1 \preceq T_\eta \preceq T_1, \quad \tau_\eta T_\eta \succeq L_h T_1 \succ G_0.$$

The positive square root in the original coefficient frame is consequently well-defined. With $D = (\tau_\eta T_\eta - G_0)^{1/2}$, put

$$R_\eta = R_0 + WD.$$

The exact equalities are

$$JR_\eta = I, \quad R_\eta^*HR_\eta = G_0 + D^*D = \tau_\eta T_\eta. \quad (C26)$$

No square root has been transported through a nonunitary congruence.

Tensor C26 and compose with beta. After the original physical source isometry A_h , the section $\mathcal{R} = (A_h R_{\eta})^{\otimes k}$ beta has value beta v and squared norm

$$\mathcal{R}^* \mathcal{R} = \tau_{\eta}^k G_{k,\eta}. \quad (C27)$$

Let X_N be exactly the scalar-polynomial physical source from $P(S)$, $\deg P \leq N$, with $S = s_1 + \dots + s_k$, and set $Y = \text{im } \mathcal{R}$. Both sources lie in the box of separate degree at most $n = \max(N, n_0)$. The section shows that their sum maps onto E . Its minimum exists and is unique modulo zero-value directions by finite-dimensional orthogonal projection. Its metric is positive definite. J2 applied before taking this minimum gives its lower bound; C27 is an admitted vector for each value and gives its upper bound. Thus

$$\Lambda_{h,n}^{-k} G_{k,1} \preceq G_N^J \preceq \tau_{\eta}^k G_{k,\eta} \preceq \tau_{\eta}^k G_{k,1}. \quad (C28)$$

For $N \geq q-1$ the scalar source is onto E , with positive minimum metric G_N ; source inclusion then gives $G_N^J \leq G_N$.

For the full box, let H_n be the one-factor polynomial Gram and J_n its remainder matrix. Its covariance is $C_n = J_n H_n^{-1} J_n^*$, and its minimum metric is $G_{\{h,n\}} = C_n^{-1}$. The box covariance is exactly

$$(J_n^{\otimes k})(H_n^{-1})^{\otimes k}(J_n^*)^{\otimes k} = C_n^{\otimes k}.$$

Its value metric is therefore $G_{\{h,n\}}^{\otimes k}$. Restriction to the exact value beta v yields

$$G_{k,n}^{\text{box}} = \beta^*(G_{h,n}^{\otimes k})\beta \preceq G_N^J, \quad (C29)$$

because every joint candidate is a box candidate. This proves J4 with the whole box kernel present. The same proof factor by factor gives the heterogeneous constants $\prod_i \Lambda_{h_i, n_i}$ and $\prod_i \tau_{\{i, \eta\}}$, with $n_i = \max(N, 2d_i - 1)$.

19.10 9. The original cross Gram and its positive-range inverse

For $N \geq q-1$ let B_X be the scalar coefficient map into the actual source, $H_X = B_X^* B_X$, and let J_N be its map onto E . Its orthogonal projector is $P_X = B_X H_X^{-1} B_X^*$. Put

$$C_X = B_X^* \mathcal{R}, \quad R_Z = (I - P_X) \mathcal{R}, \quad H_Z = R_Z^* R_Z = \Gamma - C_X^* H_X^{-1} C_X, \\ F_Z = I - J_N H_X^{-1} C_X. \quad (C30)$$

The actual value map sends R_Z to beta F_Z . R_Z lies in the same polynomial box, so J1 applied to R_Z v for every v gives

$$F_Z^* G_{k,1} F_Z \preceq \Lambda_{h,n}^k H_Z. \quad (C31)$$

In particular $\ker H_Z \subset \ker F_Z$, since $G_{\{k,1\}}$ is positive.

The sum source is the orthogonal sum $X_N + \text{im } R_Z$. Take a unitary spectral decomposition of H_Z . On each positive eigenvalue lambda its source coordinate becomes orthonormal after multiplication by $\lambda^{-1/2}$; coordinates in $\ker H_Z$ represent the zero vector and are killed by F_Z . Hence the covariance of this normal source is exactly $F_Z H_Z^{-1} F_Z^*$. The scalar covariance is $J_N H_X^{-1} J_N^* = G_N^{-1}$. Covariances of orthogonal source summands add, which proves

$$\Omega_N = F_Z H_Z^{-1} F_Z^*, \quad G_N^J = (G_N^{-1} + \Omega_N)^{-1}. \quad (C32)$$

The inverse exists because the scalar covariance is positive definite. Factoring it out gives

$$\frac{\det G_N}{\det G_N^J} = \det(I + G_N^{1/2} \Omega_N G_N^{1/2}), \quad (C33)$$

which proves J7–J8 on the original coefficient space.

The section is injective by its right-inverse property. Therefore

$$\dim \ker R_Z = \dim(Y \cap X_N), \quad \text{rank } H_Z = q - \dim(Y \cap X_N).$$

If a vector belongs to the intersection, its polynomial is both $P(s_1 + \dots + s_k)$ and of separate degree at most n_0 . Equality of physical vectors implies equality of these polynomials, because the product density is positive almost everywhere and a polynomial vanishing there is zero. If $\deg P = M$, its highest pure s_1 monomial has nonzero coefficient and degree M , so $M \leq n_0$. Consequently

$$\dim(Y \cap X_N) \leq \min(N, n_0) + 1 \leq 2d, \quad \text{rank } H_Z \geq q - 2d. \quad (\text{C34})$$

This is J9. It makes no claim that the value correction Ω_N has the same rank as H_Z .

19.11 10. Fixed restrictions, complete quotients, and the four cutoffs

Assume $2q \geq n_0$ and use the single constant $\Lambda = \Lambda_{\{h, 2q\}}$ throughout $q-1 \leq N \leq 2q$. Every relevant source lies in that same box. With $G_{\text{ref}} = G_{\{k, 1\}}$, C28 gives

$$\alpha G_{\text{ref}} \preceq G_N^J \preceq b G_{\text{ref}}, \quad \alpha = \Lambda^{-k}, \quad b = \tau_\eta^k, \quad \log(b/\alpha) = W_k = k \log(\tau_\eta \Lambda). \quad (\text{C35})$$

For any fixed injective inclusion I , congruence gives the same bounds on $I^* G_N^J I$. For any fixed onto quotient L onto its actual image, minimizing the scalar quadratic inequality over the exact affine fibre $Lv = w$ gives the same bounds on its quotient metric. That minimum is

$$Q_N = (L(G_N^J)^{-1} L^*)^{-1},$$

as follows by the minimizing vector $v = (G_N^J)^{-1} L^* Q_N w$, whose difference from any other lift is orthogonal to $\ker L$. Repeated fixed inclusions and quotients preserve these same constants, proving the assertion for every fixed subquotient.

Because Y is fixed and X_N increases, G_N^J decreases with N . Restriction and affine minimization preserve this order. If the resulting rank is r_X and $f(N) = \log \det G_{\{X, N\}}^J$, positive congruence and eigenvalue comparison imply

$$f(N_1) \geq f(N_2) \quad (N_1 \leq N_2), \quad \sup_N f(N) - \inf_N f(N) \leq r_X W_k.$$

Pair the actual four signs as

$$\mathcal{R}f = [f(q-1) - f(2q-1)] + [f(q) - f(2q)].$$

Both differences lie between zero and $r_X W_k$. This proves J11 exactly:

$$0 \leq \mathcal{R} \log \det G_{X, N}^J \leq 2r_X W_k. \quad (\text{C36})$$

Rank one includes the restriction to a specified nonzero class, with its original vector fixed across all cutoffs.

The complete finite tolerance expression is

$$W_k = k \log L_h + k \log \Lambda + 2kd_{\max} \log(1/\eta). \quad (\text{C37})$$

For the general choice $\epsilon = \min(1, t_*/(D+1))$, where D is the largest total tuple degree and $D \leq k d_{\max}$,

$$\log(1/\eta) = \log^+((D+1)/t_*) \leq \log(D+1) + \log^+(1/t_*).$$

C19 and C22 therefore give the precise J10 order

$$W_k = O_h(k \log^2(q+2) + k \log(k+2) + k \log^+(1/t_*)). \quad (\text{C38})$$

For a common-multiplicity quartet with $e=1+k(m_0-1)$ and $\text{eta}=\text{delta}/e$ when $m_0>1$, one instead reads $\log(1/\text{eta})=\log e+\log(1/\text{delta})=0_h(\log(k+2))$ directly. For simple roots $d_{\text{max}}=0$, so C37 is independent of eta and gives $W_k=0_h(k \log^2(q+2))$; on the simple quartet $q=(k+1)^2$.

For every fixed one-factor root set with $z=|Z|$, there is also the explicit polynomial growth bound

$$q \leq (kd_{\text{max}} + 1) \binom{k+z-1}{z-1}. \quad (\text{C39})$$

Indeed an unordered root-count vector is a z -tuple of nonnegative integers summing to k , so there are exactly the displayed binomial number of such vectors. Every ordered tuple has one such vector and therefore one of at most that many centers. Each center has $e_{\text{kappa}} \leq k d_{\text{max}} + 1$. Summing these inequalities proves C39, including all possible collisions. This is the polynomial growth assertion needed in J10. On the stated quartet the exact formula $q=(1+k(m_0-1))(k+1)^2$ follows because the independent real and imaginary sign counts each run from 0 through k and every pair of counts is attainable; all tuples have the same $D=k(m_0-1)$.

If a family of intermediate sources contains Y and stays inside the same box, C35 and its fixed restrictions and quotients still hold. Nesting is needed for the first inequality in C36. In the absence of nesting each of the two paired differences lies in $[-r_{\text{XW}_k}, r_{\text{XW}_k}]$, yielding the valid conclusion $|R f| \leq 2r_{\text{XW}_k}$. Common enclosing constants alone cannot fix the sign: in rank one, choosing both early metric values at alpha and both late values at b gives $R f = -2\log(b/\alpha)$. This shows exactly why the claimed intermediate-source extension needs its stated scope repair. The original joint source satisfies the nesting requirement.

19.12 11. Reading and use record

The mathematical input actually read for this derivation was:

- `COMPLETE_PROOFS(2).md`, complete Sections 1–2, equations P1–P11 and J1–J11. The initial tool output also exposed later sections; no independent verification of those later sections is asserted.
- `.../family_cost_intake_20260920/extracted/ES_RH_Family_Integration_20260920_C/proofs/01_METRIC_RETURN.md`, complete file read; only Section 1.1's explicit common-multiplicity tolerance choice was used here. Its separate imported spectral theorem was not used in the present proof.
- The same package's `proofs/02_JOINT_SOURCE.md`, bounded search excerpts around the cross-Gram and physical-source discussion. The cross-Gram identities in this file were independently proved in C30–C33; no claim of complete reading of that package is made.
- NIST DLMF, version 1.2.8, release 2026-09-15: exact formulas 5.7.6, 18.23.7, 18.19.8–9, and 25.2.10–11. These formulas were read in the rendered primary source and fetched from its original `.tex` endpoints. They supply only the classical identities identified in C1–C3, C6, and C8; all quantitative constants and all finite tensor and minimum arguments are derived above. The six unchanged TeX responses are retained in `coercivity_sources/` under their DLMF equation IDs, including 18.19.E9a.tex for the norm.

No paper absent from this list, no original arXiv source, and no entire literature corpus is represented as having been read. No PDF was used. No remote publication or graphical rendering was performed in this independent derivation subtask.

Complete preceding joint-minimum receiver

The complete supplied proof [1], reproduced with explicit scope corrections in this edition, evaluates the metric change caused by admitting its tensor section into the original scalar source. This receiving calculation proves its consequences for the constructed section itself, all intermediate original polynomial sources, the original observation, and the previously established canonical heat calculation. The independent source estimate uses Apostol's zeta formulas, Askey–Roy's Gamma formulas and the orthogonal-polynomial formulas of Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [2, 3, 4]. All original masses are retained.

20 Objects and the fully evaluated mixed minimum

Fix the programme's actual hypothetical off-line quartet with $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, common complete multiplicity m_0 , and complete divisor h . Put

$$e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad k \equiv 1 \pmod{4}, \quad q - 1 \leq N \leq 2q. \quad (\text{JM1})$$

All limits fix the quartet, m_0 , its positive jet weights and its one-factor section. The original cyclic value space is $E = \mathbb{C}[S]/(\chi_k)$, with every multiplicity e at every sum centre. The physical source consists of the original product of $w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$. X_N contains the original polynomials $P(S)$ of degree at most N . The fixed tensor section $R = \mathcal{R}_{k,\eta}$ has separate degree at most $2\deg h - 1$ and full tensor value βv . The cyclic metric G_N is the minimum over X_N ; G_N^J is the minimum over $X_N + \text{im } R$, with the same full value. The section's own Gram is $G^S = \tau_\eta^k G_{k,\eta}$.

In the original source Gram, let B_X be the scalar inclusion, $H_X = B_X^* B_X$, J_N its cyclic value, and $C_X = B_X^* R$. Then the complete normal source and its value are

$$H_Z = R^* R - C_X^* H_X^{-1} C_X, \quad F_Z = I - J_N H_X^{-1} C_X, \quad \Omega_N = F_Z H_Z^+ F_Z^*. \quad (\text{JM2})$$

The star uses the original physical form. The inverse H_Z^+ acts on its whole positive range. Physical equality gives $\ker H_Z \subseteq \ker F_Z$. Orthogonally decomposing each candidate into X_N and its normal component, and minimizing the two positive quadratic forms under their combined value constraint, gives

$$G_N^J = (G_N^{-1} + \Omega_N)^{-1}. \quad (\text{JM3})$$

This is the exact map from the mixed source to its minimum, also proved in GF18–32 and MA1–12 [6, 7].

The incoming P1–P11 proves, on the entire polynomial space of degree at most n , a positive form comparison with the unscaled Gamma measure $\sigma = |\Gamma(1/4 + iy/2)|^2/(2\pi)$ and a complete jet bound

$$a_{h,n} \|P\|_\sigma^2 \leq \|P\|_{w_h}^2 \leq b_{h,n} \|P\|_\sigma^2, \quad \|J_h P\|_{T_1}^2 \leq \Lambda_{h,n} \|P\|_{w_h}^2, \quad \log \Lambda_{h,n} = O_h(\log^2(n + 2)). \quad (\text{JM4})$$

The complete finite constants and derivation are retained in P1–P11. For the disk step, the mean-value and Poisson formulas [5] give its constants directly. If $f(0) \neq 0$ and r meets no zero, factor out all zeros a in $|z| < r$. The logarithm of the remaining zero-free modulus is harmonic, and the circle mean of $\log |re^{it} - a|$ equals $\log r$. Thus the circle mean of $\log |f|$ minus $\log |f(0)|$ equals $\sum_{|a| < r} \log(r/|a|)$. Letting r increase to four proves the zero count in P6. For the nonnegative harmonic function $u = \log(M_n/|F|)$ on the radius-three disk, the Poisson kernel on an inner circle of radius r is at most $(r + 2)/(r - 2)$ on $|z| \leq 2$. Its circle mean is $u(0)$; letting r increase to three gives $u(z) \leq 5u(0)$. This proves the exact factor five in P7, including possible zeros on the outer circles. Tensoring the

positive matrix inequality, before any minimization, applies to every mixed polynomial. Hence, with $\Lambda = \Lambda_{h,2q}$,

$$F_Z^* G_{k,1} F_Z \preceq \Lambda^k H_Z, \quad \Lambda^{-k} G_{k,1} \preceq G_N^J \preceq G^S \preceq \tau_\eta^k G_{k,1}. \quad (\text{JM5})$$

Thus small normal-source eigenvalues are controlled together with their actual value directions. No separate bound on H_Z^{-1} is assumed. The normal rank is at least $q - 2 \deg h$: a scalar polynomial lying in the tensor section has separate degree at most $2 \deg h - 1$; its highest pure monomial then forces its total scalar degree to be at most that number. Injectivity of the physical polynomial map turns this into the stated dimension bound.

21 The same full spectrum for the section and for intermediate sources

Use the explicitly transported common coefficient coordinate $x = (S - c)/(iq)$, and let $D_k = \text{diag}((k/q)^n)_{n=0}^{q-1}$. The finite coefficient and Hermite estimates S2–S8 give

$$A_- I \preceq G_N \preceq A_+ I, \quad \nu_- D_k^2 \preceq G_{k,1} \preceq \nu_+ D_k^2, \quad |\log A_\pm| + |\log \nu_\pm| = O_h(q). \quad (\text{JM6})$$

Their proof retains the whole lattice product of root differences and all divided derivatives. In particular the derivative row at order j has weight $W_{ab}(j!)^2 \binom{e-1}{j} k^{-2j}$; this physical Jacobian is essential.

Here and below $\eta = 1$ for $e = 1$, and $\eta = \delta/e$ for $e > 1$, or more generally $\log(1/\eta) = O_h(\log(k+2))$. Since each jet order is at most $e - 1$,

$$\eta^{2(e-1)} G_{k,1} \preceq G_{k,\eta} \preceq G_{k,1}. \quad (\text{JM7})$$

Therefore the generalized eigenvalues λ_j^S of (G_N, G^S) obey the following additional finite comparison:

$$\frac{A_-}{\tau_\eta^k \nu_+} (q/k)^{2(j-1)} \leq \lambda_j^S \leq \frac{A_+}{\tau_\eta^k \eta^{2(e-1)} \nu_-} (q/k)^{2(j-1)}. \quad (\text{JM8})$$

To prove this, divide the quadratic-form inequalities JM6–JM7, and apply the minimum–maximum characterization to the same coefficient subspaces. The ordered eigenvalues of D_k^{-2} are $(q/k)^{2(j-1)}$. Taking logarithms gives

$$\log \lambda_j^S = 2(j-1) \log(q/k) + O_h(q), \quad \log \text{cond}(G_N, G^S) = 2(q-1) \log(q/k) + O_h(q). \quad (\text{JM9})$$

These section-relative eigenvalues need not be at least one. The joint-relative eigenvalues are at least one because $G_N^J \preceq G_N$. Both comparisons use the literal identity on the original E .

More generally take any physical polynomial source V_N containing $X_N + \text{im } R$ and contained in the complete separate-degree $2q$ box. Impose the original full tensor value βv . The same pre-minimum bound and the same section candidate prove

$$\Lambda^{-k} G_{k,1} \preceq G_{V,N} \preceq \tau_\eta^k G_{k,1}. \quad (\text{JM10})$$

Consequently every such source has exactly the logarithmic spectrum S1, the condition-number asymptotic S10, and the simple-quartet determinant asymptotic V13, with the same error order. This proves uniformity over these sources, rather than identifying their finite minimizers. For two such sources, putting $W = k \log(\tau_\eta \Lambda)$,

$$e^{-W} G_{V_1,N} \preceq G_{V_2,N} \preceq e^W G_{V_1,N}, \quad |\log \det G_{V_2,N} - \log \det G_{V_1,N}| \leq qW. \quad (\text{JM11})$$

For the second assertion use the two common enclosures directly; each determinant lies in an interval of length qW around the same reference determinant. It need not be charged twice.

22 Observed spectrum, determinant and tolerance

Let $\Lambda_O : E \rightarrow B$ be the original onto observation, with its original period, branch and physical unit. Write

$$Q = (\Lambda_O G_N^{-1} \Lambda_O^*)^{-1}, \quad Q^J = (\Lambda_O (G_N^{-1} + \Omega_N) \Lambda_O^*)^{-1}, \quad S_O = Q^{1/2} \Lambda_O G_N^{-1/2}. \quad (\text{JM12})$$

Then $S_O S_O^* = I_B$ and the observed relative covariance is the exact compression of $I + G_N^{1/2} \Omega_N G_N^{1/2}$ by S_O . The minimum–maximum characterization proves $\lambda_j \leq \mu_j \leq \lambda_{j+m}$, $m = \dim \ker \Lambda_O$. On the original simple five-orbit domain $m = 8k - 16$ this retains the whole limiting logarithmic spectrum. Factoring each positive metric along the same kernel gives, without an invariant-kernel assumption,

$$\Delta_N = \Delta_{K,N} + \Delta_{B,N}, \quad 0 \leq \Delta_{K,N} \leq m[2(q-1) \log(q/k) + E_{h,k}]. \quad (\text{JM13})$$

Indeed interlacing bounds the quotient determinant below by the product of the $q - m$ smallest full relative eigenvalues; positivity of the kernel comparison supplies the other inequality. Thus the full V13–V14 result passes to this original observation:

$$\Delta_{B,N} = q^2 [\log(q/k) - \tfrac{3}{2} - E_{\square} - 2\mathfrak{F}((N+1-q)/q)] + O_h(kq \log^2(q+2)), \quad (\text{JM14})$$

$$\mathcal{R} \Delta_{B,N} = C_B q^2 + O_h(kq \log^2(q+2)). \quad (\text{JM15})$$

Here $\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}$. The complete rectangle energy V9, all ordered-tuple weights, and the original FW monic-profile theorem are retained in the proof.

Two scope points are made explicit. For a fixed rank- r restriction or attained quotient of JM10, the common enclosure gives $|\mathcal{R} \log \det G_{V,X,N}| \leq 2rW$. Its lower bound is zero only when the four sources are nested in increasing N , as the actual $X_N + \text{im } R$ sources are. This nesting was implicit in the incoming extension of J11 to intermediate sources.

For the observed terminal class $w = \Lambda_O v_{\lambda} \neq 0$, retain $\theta_N = \|P_{K,N} v_{\lambda}\|_{G_N}^2 / \|v_{\lambda}\|_{G_N}^2$. The exact identity and finite error statement are

$$\begin{aligned} \mathcal{R} \log \frac{w^* Q w}{w^* Q^J w} &= C_{\partial} q + \mathcal{R} \log(1 - \theta_N) + E_{\text{FW}} - E_J, \\ |E_{\text{FW}}| &= O_h(k + \log q), \quad 0 \leq E_J \leq 2k \log(\tau_{\eta} \Lambda). \end{aligned} \quad (\text{JM16})$$

This follows by substituting $w^* Q w = (1 - \theta_N) v_{\lambda}^* G_N v_{\lambda}$ and applying J11 to the rank-one observed minimum. The smaller error $O_h(k \log^2(q+2))$ therefore holds at the declared polynomial tolerances. It is not claimed uniformly for arbitrarily small η . For the unobserved terminal class the cancellation $\tau_{\eta}^k \eta^{2k(m_0-1)} = L_h^k$ proves the stronger all- η bound O9a–O10 exactly as written in the supplied proof.

23 Carry the result back into the original heat calculation

Let $A = M_S$ on E , and retain both minimum sections $L = G_N^{-1} \Lambda_O^* Q$ and $L^J = (G_N^{-1} + \Omega_N) \Lambda_O^* Q^J$. Put $P_K = I - L \Lambda_O$. Subtracting the sections and using $Q - Q^J = Q \Lambda_O \Omega_N \Lambda_O^* Q^J$ gives

$$L^J - L = P_K \Omega_N \Lambda_O^* Q^J, \quad A_B^J - A_B = \Lambda_O A P_K \Omega_N \Lambda_O^* Q^J. \quad (\text{JM17})$$

This retains the actual kernel leakage and connects the two actions exactly. No identification of their Hilbert-space adjoints is made.

For a metric H and its actual observed action put $M_H = (A_{B,H} - cI)/i$ and

$$d_{H,N}(t) = \text{Re Tr } e^{-tM_H^2} - \text{Tr } e^{-tM_H^{\dagger H} M_H}.$$

In $G_{k,1}$ the complete nilpotent chain has norm at most $(e+1)/2$, and its centres have modulus at most kR_0 . Let $C_k = kR_0 + (e+1)/2$ and $U_k = e^{W/2}C_k$. The quotient map is a contraction and its minimum section is an isometry; JM5 therefore gives $\|M_{Q^J}\| \leq U_k$. The exponential series and positivity give the finite estimate

$$|d_{Q^J,N}(t)| \leq r_B \{tU_k^2 e^{tU_k^2} + tU_k^2\} \leq 2r_B tU_k^2 e^{tU_k^2}. \quad (\text{JM18})$$

This bound uses the centred operator directly, so no centre has been subtracted from a norm estimate after the fact. Since $W = O_h(k \log^2(q+2)) = o(q)$, at $t_k = e^{-2bq}$, $b > 0$, its right side is $\exp(-2bq + o_h(q))$. It is uniform in the full cutoff window.

On the original simple-packet and period domain of OH16 [8], retain $a_0 = \psi(0)$, $a_1 = \psi(1)$ and $b_* = (a_0 + \max\{a_1, a_0/2\})/2$. That theorem proves $\mathcal{R}d_{Q,N}(e^{-2b_*q}) = 2 + \varepsilon_{\text{OH}}(k)$ with $\varepsilon_{\text{OH}}(k) \rightarrow 0$ and its stated finite guards. Combining it with the four actual JM18 bounds proves the new return

$$\boxed{\mathcal{R}[d_{Q,N}(e^{-2b_*q}) - d_{Q^J,N}(e^{-2b_*q})] \longrightarrow 2.} \quad (\text{JM19})$$

More precisely the distance from two is at most $|\varepsilon_{\text{OH}}(k)| + 8r_B t_k U_k^2 e^{t_k U_k^2}$. The heat traces are compared at the same physical time, on the same observed value space, through the exact action map JM17. The metric-correction heat S13 acts instead on $G_N^{1/2} \Omega_N G_N^{1/2}$ at scale $e^{-2bq \log(q/k)}$; JM17 is the proved connection to arithmetic multiplication, and no equality of these two operators is asserted.

24 The original kernel return through its entire Gamma minimum

There is a stronger use of the new form bound at the unresolved kq scale. This section keeps the actual simple-packet kernel $K = \ker \Lambda_O$, of dimension $m_K = 8k - 16$, in a fixed frame I_K . It does not replace that frame by a coordinate truncation or by the larger conductor ideal. Put $n = 2q$, $\ell = (k-1)/4$, and

$$b_j = 2j + \frac{1}{2}, \quad c_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad u_j = 2(n+j+1) + b_j \quad (0 \leq j < \ell). \quad (\text{JM21})$$

Empty products equal one. Here is the full convolution constant. With $\widehat{f}(t) = \int e^{-ity} f(y) dy$, the beta integral gives

$$\frac{\Gamma(a+ix)\Gamma(a-ix)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{ixv} (2 \cosh(v/2))^{-2a} dv, \quad a > 0.$$

Indeed the product of the two Euler integrals, under $(u,v) = (r\theta, r(1-\theta))$, has Jacobian r . The r integral gives $\Gamma(2a)$; setting $z = \log(\theta/(1-\theta))$ in the remaining beta integral gives the displayed formula. The function on the right and its first two derivatives are integrable. Two integrations by parts make its Fourier transform integrable, so Fourier inversion and $y = 2x$ give

$$\int e^{-ity} \frac{|\Gamma(a+iy/2)|^2}{2\pi} dy = \frac{2\Gamma(2a)}{(2 \cosh t)^{2a}}.$$

Thus $\widehat{\sigma}(t) = \sqrt{2\pi}(\cosh t)^{-1/2}$ and

$$\sigma^{*k}(y) = \frac{(2\pi)^{k/2} 2^{k/2-1}}{\Gamma(k/2)} \frac{|\Gamma(k/4+iy/2)|^2}{2\pi} = c_k \prod_{j < \ell} (y^2 + b_j^2) \sigma(y).$$

The last equality is the Gamma recurrence; the mass is exactly $(2\pi)^{k/2}$. The physical variable remains $S = c + iy$, $c = k/2$. Below $\|P\|_\sigma$ means $\|P(c + i \cdot)\|_\sigma$ and likewise for the convolution norm. For every polynomial P of degree at most n , the tensorized P2 inequality and the Gamma multiplication estimate in the proof of P5 give

$$\alpha_k \|P\|_\sigma^2 \leq \|P\|_{w_h^{*k}}^2 \leq \beta_k \|P\|_\sigma^2, \quad \alpha_k = a_{h,n}^k c_k \prod_{j < \ell} b_j^2, \quad \beta_k = b_{h,n}^k c_k \prod_{j < \ell} u_j^2. \quad (\text{JM22})$$

For the lower inequality each $y^2 + b_j^2 \geq b_j^2$ on the real line. For the upper one, write the factor as the squared modulus of $\prod_{j < \ell} (y + ib_j)$. After j factors the degree is at most $n + j$, so that multiplication estimate and the triangle inequality bound multiplication by its next factor by u_j . Tensorization applies to $P(\sum y_i)$ because its separate degrees are at most n ; the convolution identity then gives JM22. Thus the comparison holds on the complete source, including every relation candidate.

Define the explicit nonnegative logarithmic width

$$L_k^{\text{form}} = \log(\beta_k/\alpha_k) = k \log(b_{h,n}/a_{h,n}) + 2 \sum_{j < \ell} \log(u_j/b_j) = O_h(k \log^2(q+2)). \quad (\text{JM23})$$

The fixed mass c_k is present in both finite bounds before it cancels from this ratio. Let G_N^σ be the full minimum of this order-one Gamma norm under the unchanged original remainder map modulo $\chi_k(S)$, with $S = c + iy$. Taking the identical affine minimum in JM22 proves $\alpha_k G_N^\sigma \preceq G_N \preceq \beta_k G_N^\sigma$. Restricting by the original I_K and taking determinants gives

$$m_K \log \alpha_k \leq \log \det H_{K,N} - \log \det H_{K,N}^\sigma \leq m_K \log \beta_k, \quad H_{K,N}^\sigma = I_K^* G_N^\sigma I_K. \quad (\text{JM24})$$

The four cutoffs share these same bounds. Pairing their signs therefore proves the finite and asymptotic original-kernel return

$$\left| \mathcal{R} \log \det H_{K,N} - \mathcal{R} \log \det H_{K,N}^\sigma \right| \leq 2m_K L_k^{\text{form}} = O_h(k^2 \log^2(q+2)) = o_h(kq). \quad (\text{JM25})$$

Every lower-source relation was minimized before restriction. The inequality concerns the original kernel at all four cutoffs.

The Gamma metric in JM25 is the following explicit finite matrix. Set $Q(y) = i^{-q} \chi_k(c + iy)$ and $T : \mathbb{C}[S]/(\chi_k) \rightarrow \mathbb{C}[y]/(Q)$, $T[P] = [P(c + iy)]$. Its inverse substitutes $y = (S - c)/i$; its coefficient matrix is triangular with diagonal $1, i, \dots, i^{q-1}$. The transported frame is $I_{K,y} = T I_K$, the metric is $(T^{-1})^* G_N^\sigma T^{-1}$, and their restriction is exactly $I_K^* G_N^\sigma I_K$. Thus every root, multiplicity and frame factor is retained. In the following y -coordinate formulas write I_K for this explicitly transported frame. Set

$$p_0 = 1, \quad p_1 = y, \quad p_{j+1} = y p_j - j(j - \tfrac{1}{2}) p_{j-1}, \quad v_j = \text{rem}_Q p_j, \quad \gamma_j = \sqrt{2\pi} j! (\tfrac{1}{2})_j. \quad (\text{JM26})$$

Orthogonality from [4] gives exactly

$$C_N^\sigma = \sum_{j=0}^N \frac{v_j v_j^*}{\gamma_j}, \quad H_{K,N}^\sigma = I_K^* (C_N^\sigma)^{-1} I_K. \quad (\text{JM27})$$

The first q monic remainder columns form an invertible triangular matrix, so $C_N^\sigma > 0$ for every $N \geq q - 1$. The covariance formula follows by mapping the full orthogonal source basis to its quotient and then minimizing. Thus JM27 includes all roots, multiplicities and earlier relations. It does not replace the inverse followed by restriction with the inverse of a restriction.

For the change from the original to the joint kernel metric, JM5 and nesting also give $0 \leq \mathcal{R} \log \det H_{K,N}^J \leq 2m_K W$. Combining this with JM25 proves

$$\mathcal{R} \log \det H_{K,N}^\sigma - 2m_K(L_k^{\text{form}} + W) \leq \mathcal{R} \Delta_{K,N} \leq \mathcal{R} \log \det H_{K,N}^\sigma + 2m_K L_k^{\text{form}}. \quad (\text{JM28})$$

Both errors are $o_h(kq)$ at the declared polynomial tolerances. On this simple-divisor domain $d_{\max} = 0$, so $\tau_\eta = L_h$ and $W = k \log(L_h \Lambda_{h,2q})$ is independent of η . Thus here the same bound holds for every $0 < \eta \leq 1$ with its section fixed across the four cutoffs. Therefore the next kernel calculation is the large-degree determinant in JM27 with the actual conductor-defined frame I_K . The reduction has the precision to preserve its kq coefficient; it does not yet supply that coefficient.

25 Backward propagation and the next actual calculation

The earlier CR4–12 comparison lower bounds remain valid, but JM9 now supplies the larger complete leading condition-number value for their constructed-section metric. GF18–32, MT10–15 and MA1–12 remain the exact finite maps. P1–P11/JM5 supply their formerly missing whole-source coercivity, S1 evaluates the relative spectrum, and V13/JM14 evaluate the full and observed determinants through their q^2 coefficients. JM19 carries these results back through the original canonical heat result rather than overwriting it.

The actual kernel’s finer kq coefficient now reduces, with $o_h(kq)$ error, to the explicit full Gamma minimum JM27. That determinant’s asymptotic coefficient, the independent proper source minimum and the consecutive projected-current phase remain unevaluated. The latter quantities retain the actual mixed moment matrix MA3–MA4, retaining

$$B_X^*(H_a - H_0)Z_0, \quad Z_0^*(H_a - H_0)Z_0 - E^*(B_X^* H_a B_X)^{-1} E. \quad (\text{JM20})$$

The large determinant comparison does not allocate its finer terms between those original restrictions. The new bound controls their joint value directions and leaves the original maps available for that further calculation.

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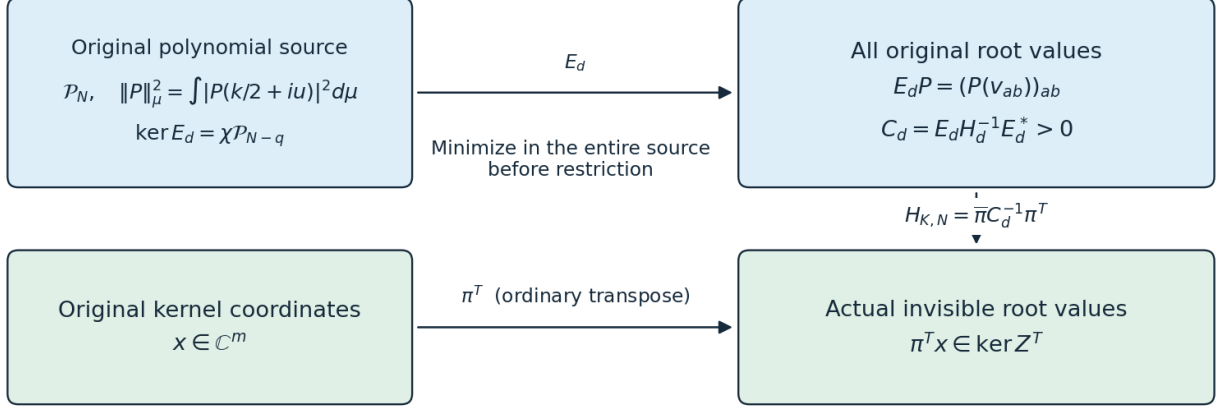
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- [7] Split-Zero research programme, *The complete arithmetic correction to the tensor-source projection*, MA1–12, result055.
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The exact determinant maps and their programme use

The original kernel, with every source relation retained

Exact finite maps: $S = k/2 + iu$, $q = (k + 1)^2$, $m = 8k - 16$, $r = q - m$, $d = N + 1$.

Original simple-packet five-orbit domain; $k \geq 9$, $k = 1 \pmod{4}$, $N \geq q - 1$.



$Z = [M_A, S_k Z_k]$: the whole invariant image, not only the conductor columns M_A .

$$\det H_{K,N} = g_{\pi,Z} \frac{\det(Z^T C_d \bar{Z})}{\det C_d}$$

$$= \frac{g_{\pi,Z}}{|V_\chi|^2} \frac{\sum_{|I|=|J|=r} z_I \bar{z}_J V_I \bar{V}_J B_{N+1-r}(\chi_I, \chi_J)}{B_{N+1-q}(\chi, \chi)}$$

Every $I \neq J$ cross term remains. The numerator size is $n + m$, not $n = N + 1 - q$.

Gamma receiver: $|\mathcal{R} \log \det H_K - \mathcal{R} \log \det H_K^g| \leq 2mL_k^{\text{form}} = o(kq)$.

$\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}$. The kq coefficient is not evaluated by this diagram.

Proof: MR8-23. Human source: Akemann-Vernizzi, hep-th/0212051v2, Th and deg.

Original programme inputs: OCF1-24, Toda (11)-(16), JM21-28. This is an exact-map schematic.

The diagram records MR1-23, not sampled arithmetic data. Its coefficient frame contains both the full conductor image and the remaining invariant directions. The dual map uses an ordinary transpose; the Hilbert Gram uses adjoints. The positive integral retains every mixed cross term and the factorial. Its numerator degree is $n + m$, while its denominator degree is n . The preceding Gamma comparison preserves the sought kq coefficient of the four-cutoff return. The exact finite expression is proved here; that coefficient has not yet been evaluated. The human source of the finite characteristic-polynomial identity is G. Akemann and G. Vernizzi, Nuclear Physics B 660 (2003), 532-556, arXiv:hep-th/0212051v2, original labels Th, Ddef, proof1 and deg. The complete source was read through its bibliography. The original Toda quotient remains credited to its preceding programme source, retained in full in the provider directory.